

Prioritising low-risk and high-potential circular economy strategies for decarbonisation: A meta-analysis on consumer-oriented product-service systems

R. Koide^{a,b,c,*}, S. Murakami^b, K. Nansai^a

^a Material Cycles Division, National Institute for Environmental Studies, Onogawa 16-2, Tsukuba, Ibaraki, 305-8506, Japan

^b Graduate School of Engineering, The University of Tokyo, Hongo 7-3-1, Bunkyo-ku, Tokyo, 113-8656, Japan

^c Institute for Global Environmental Strategies, Kamiyamaguchi 2108-11, Hayama, Kanagawa, 240-0115, Japan

ARTICLE INFO

Keywords:

Product service systems
Circular economy
Greenhouse gas emissions
Life cycle assessment
Consumer product
Systematic review
Meta-analysis

ABSTRACT

The circular economy has potential synergies with climate change mitigation, although quantitative evidence for these synergies in the consumer product sector has not yet been systematically evaluated. We conducted a systematic review of existing studies of consumer-oriented product-service systems (PSS) on the lifecycle impacts, focusing on a meta-analysis of global warming potentials. Potential reductions in greenhouse gas emissions and the risk of backfire as a result of 10 types of circular economy strategies were investigated by reviewing 103 studies and more than 1500 scenarios for consumer durable and semi-durable products. As many as three-quarters of the reviewed studies analyse only one circular economy strategy rather than comparing (e.g., reusing, sharing, and renting) or integrating multiple strategies (e.g., leasing combined with remanufacturing). Approximately one-third of the reviewed studies rely on assumptions for set parameters of use, transport, and end-of-life phases, whereas more than half of the studies do not consider service and infrastructure provisions, or imperfect product substitution. Among the reviewed case studies, upgrading, repair, refurbishing, and pooling showed moderate to high improvement potentials with lower risks of backfiring, whereas servitisation, sharing, and reuse were associated with higher backfire risks, but they had high improvement potentials. The risk of backfiring was explained by factors such as transport, number of uses, product substitution and lifetime, maintenance, energy source, and energy efficiency. This meta-analysis revealed the importance of prioritising low-risk strategies, controlling for rebound effects of high-potential strategies, and integrating multiple strategies in order to utilize PSS for climate change mitigation.

1. Introduction

In recent years, significant changes have occurred in consumption and production patterns. There has been a shift in the dominant logic of businesses from focusing on tangible products to service provision [1], as illustrated by the rise of a sharing economy [2] and a digital economy [3]. The term product-service systems (PSS) refers to “a mix of tangible products and intangible services designed and combined so that they are jointly capable of fulfilling final customer needs” (page 1552) [4]. PSS covers a wide range of practices, which include shifting from product sales through to use on sharing platforms and leasing; extending product lifetime through repair, upgrading, reuse, and remanufacturing; and substituting goods with services [5,6], which can in turn be classified as being result-oriented, use-oriented, and product-oriented PSS [7]. The

PSS can be, if intentionally designed to be geared towards sustainability, a tool for realising the transition from a linear economy to a circular economy (CE) [8]. In addressing environmental sustainability, policies for shifting the current practices of manufacturing industries and for reducing excessive natural resource consumption are necessary [9]. CE is among the most promising approaches for sustainable development through addressing the constraints of the linear patterns of production, consumption, and disposal [10]. A CE incorporates a variety of strategies for the circular use of products and resources, ranging from product longevity, intensification of use, to prevention [11]. Recent policy developments in the field of CE, such as the European Union (EU) Action Plan for 2020 [12], proposes the establishment of sustainability principles incorporating product durability, reusability, upgradability, reparability, remanufacturing, and product-as-a-service to guide the development of policies and legislation.

* Corresponding author. Material Cycles Division, National Institute for Environmental Studies, Onogawa 16-2, Tsukuba, Ibaraki, 305-8506, Japan.
E-mail address: koide.ryu@nies.go.jp (R. Koide).

<https://doi.org/10.1016/j.rser.2021.111858>

Received 4 August 2021; Received in revised form 8 October 2021; Accepted 31 October 2021

Available online 24 November 2021

1364-0321/© 2021 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Abbreviations

B2C	Business to Consumer
C2C	Consumer to Consumer
CE	Circular economy
CED	Cumulative energy demand
EoL	End-of-life
EU	European Union
GHG	Greenhouse gas
LCA	Life cycle assessment
ICT	Information and communication technology
IQR	Interquartile range
MFA	Material flow analysis
P2P	Peer to Peer
PSS	Product-service systems

To achieve the transition to a carbon neutral society envisaged by the Paris Agreement, consideration of the product lifecycle perspective is necessary [13]. In this context, the application of CE principles in various industrial sectors may have substantial mitigation potentials, including through reductions in embodied energy [14]. A transition to a CE could potentially be an effective means for climate change mitigation, considering that 45% of greenhouse gas (GHG) emissions are directly linked to the supply of daily-use products [15]. From a life cycle perspective, household consumption is the largest final demand category in carbon footprints; of this consumption, 37% is attributed to manufactured products, clothing, and services in the EU [16]. At present, the dominant linear types of business models in the consumer goods sector envisage considerable potential in mainstreaming the CE [17], and Business-to-Consumer (B2C) PSS have been widely discussed in terms of both business model acceptability and sustainability [18].

Previous studies have concluded that some PSS, such as renting, sharing, pooling, and functional results, can potentially lead to high gains in environmental potentials [7]. However, the expected environmental benefits from PSS are not always clear [5,7]. The offset in the anticipated environmental benefits due to systemic effects are referred as rebound effects (partly reduced benefits) and backfire effects (net negative impacts) [19]. CE initiatives are to be well-designed with regular evaluation, including assessment of value chains, material flows, and products [20]. To this end, life cycle assessment (LCA) can be a useful tool for evaluating the sustainability impacts of PSS, including the consideration of rebound effects [21]. Although numerous LCA case studies on PSS have been published to date [18,22], individual studies or qualitative syntheses have not been able to comprehensively evaluate the potential contributions to climate change mitigation and risks of backfire from various types of CE strategies.

Published reviews of the sustainability impacts from PSS are typically limited in scope, particularly with respect to the application of quantitative review methods. A previous review study, which examined the advantages and disadvantages of PSS in terms of environmental sustainability [18], was based on a qualitative review of literature and did not provide any quantitative synthesis of the level of expected impacts or the reasons for backfire effects. In addition, some review studies have focused only on concepts and definitions [23], specific sector analyses such as textiles [24] and manufacturing [25], and specific CE strategies such as upgrading [26]. However, these studies do not provide a cross-cutting assessment of various CE strategies and products, and thus, are not well suited for comparing multiple CE strategies or for examining the potential impacts resulting from the combination of such strategies. Most of these reviews have been qualitative in nature, and relatively few have undertaken quantitative synthesis of results. Quantitative meta-analysis can combine and reanalyse data from existing studies to generate new findings, and such an approach has been applied to LCA

studies [27–29]. However, previous meta-analyses related to PSS have been limited to general assessments of CE [14], reuse [30], and household-level mitigation actions [31], and have not examined or compared a variety of concrete CE strategies and consumer products. Thus, a research gap related to the sustainability impacts of PSS is that no studies have conducted quantitative syntheses of GHG reduction potentials and backfire risks of PSS for various consumer products and CE strategies.

Applying LCA to PSS has numerous challenges, and further methodological improvements are expected in order to more accurately assess and design sustainable PSS. A previous review study focused on the application of LCA methods to PSS [22]; however, the study only provided an abstract overview and did not comprehensively propose specific points for methodological improvement. A series of previous studies discussed challenges in applying LCA to PSS and proposed the adoption of step-wise guidelines [6,21,32]. However, this proposal was based mainly on expert consultation and reviews of existing guidelines. Thus, another research gap related to methodology is that no studies have examined the shortcomings of current practices and the necessary improvements that are required for optimizing assessment methods based on the extensive collection of existing case studies.

Given these limitations in the existing literature, systematic reviews of mitigation potentials and methodological approaches used for the assessment of consumer-oriented PSS could contribute to leveraging CE in the consumer product sector to mitigate climate change. This review aims to address two key research questions related to consumer-oriented PSS. The first research question is *what types of CE strategies employing PSS can reduce GHG emissions through the life cycle of durable and semi-durable consumer products?* To answer this question, a qualitative synthesis (meta-analysis) focusing on improvement potentials in greenhouse gas emissions by PSS and the risk of backfire in terms of both probability and severeness were conducted. The second question is *what kind of methodological improvements are necessary in the assessment of life cycle impacts of consumer-oriented PSS?* To address this question, a systematic review of studies on life cycle impacts of PSS was conducted, focusing on their methodical approaches. The key components of this review study can therefore be summarised as follows:

- **Population (P):** Durable/semi-durable consumer products
- **Intervention (I):** Introduction of CE strategies through PSS
- **Comparator (C):** Life cycle GHG emissions
- **Outcome (O):** The improvement rate of global warming potentials per functional unit
- **Study types:** Quantification of global warming potentials through life cycle thinking, including LCA, simplified or screening LCA, input-output LCA, and simulation studies

2. Material and methods

Following the guidelines set out in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [33], this study was conducted using the following steps: i) literature search and eligibility check, ii) qualitative synthesis, and iii) quantitative synthesis (meta-analysis).

2.1. Literature survey and screening

Literature for review was identified using a title, abstract, and keyword search of Web of Science Core Collection and Scopus. Consistent with the key components of research questions, the following three components were included as search terms. Although the focus of the meta-analysis (quantitative synthesis) is on GHG emissions, other types of life cycle impacts were also included in the search terms for the purpose of conducting qualitative synthesis of methodological approaches used for evaluating PSS. Details of search terms are included in section 1.1 in the supporting information 1 (hereinafter, SI 1).

- **CE strategies:** servitisation, subscription, sharing, renting, leasing, repair, reuse, remanufacturing, refurbishing, refilling, upgrading, upcycling, durability, longevity, modularity, collaborative consumption, product service system
- **Evaluation of life cycle impacts:** LCA (or life cycle impact, energy, resource, emission, costing, simulation), environmental potential, global warming potential, cumulative energy demand
- **Consumer products:** consumer, household, electric, electronic, b2c, c2c, p2p goods (or product, equipment, tool), vehicle, car, bicycle, textile, clothes, appliance, electronics, furniture, lighting, book

As eligibility criteria, studies focusing on at least one CE strategy through the introduction of PSS were included, while studies focusing merely on improving products or processes, substituting materials, recycling materials, or shifting consumer behaviours without PSS were excluded. The products considered in this review are durable and semi-durable products targeting consumers, such as home appliances, information and communication (ICT) equipment, vehicles, and clothes, including reusable products shifted from single-use products; however, real estate and related areas were excluded. Peer-reviewed articles, conference proceedings, and book chapters published in English that quantify global warming potentials (for meta-analysis) and other types of impacts (for qualitative review) with any geographical boundaries were included. Further details of the eligibility criteria used for inclusion in the review are described in section 1.2 in SI 1.

A literature search was conducted on April 10, 2021. In total, 1608 studies were identified by a keyword search (662 in Web of Science and 946 in Scopus). After removing duplicates, 1160 studies were selected for screening. After title and abstract screening, 199 studies were selected for full-text eligibility checks. The full texts of these articles were checked and collated based on the eligibility criteria, and 103 articles were selected for qualitative synthesis at the literature level.¹ Of these articles, 65 examined comparable global warming potentials through the introduction of CE strategies; these were used for quantitative synthesis (meta-analysis) at the scenario level. The remaining 38 articles, including quantitative studies on life cycle impacts other than GHG (20 articles) and studies on GHG that were lacking in comparable results from baselines (18 articles), were used only for the literature-level analysis.

2.2. Qualitative synthesis of methodological approaches

The systematic review was conducted at the literature level and the scenario level. The literature-level analysis was conducted to clarify existing methodological approaches for assessing consumer-oriented PSS and to draw recommendations for further improvements, focusing on the following: i) methodological approaches, ii) target product-services and CE strategies, iii) reasons for deviations of global warming potentials, and iv) use of empirical data. First, methodological approaches in each paper, i.e., the methodologies used for quantifying life cycle impacts, additional methodologies, objectives, functional units, indicators, and types of parameters used in the scenario and sensitivity analyses, were coded. These data were analysed based on the proportion of studies that adopted a specific approach, and details of the methodological approaches were investigated based on the descriptions of the reviewed studies. Second, target product-services and CE strategies were also coded, and these were visualised as a parallel categories diagram to examine the combination of product-service types, CE strategies, and PSS types. The following 10 types of CE strategies were considered in this study: servitisation (result-oriented), pooling, Consumer to Consumer (C2C) sharing, renting, leasing (use-oriented), reuse, repairing, upgrading/modularity, refurbishing/remanufacturing, and durability (product-oriented) (PSS types are shown in parenthesis). Here, the

comparison between CE strategies (e.g., comparisons between renting and sharing of a particular product) and the combination of multiple strategies (e.g., the combination of leasing and remanufacturing) were distinguished and analysed using a network graph to identify the CE strategies that were compared and combined in the reviewed studies.

Third, the conclusions of the reviewed studies, in terms of the possibility of backfire effects (e.g., backfire expected, backfire not expected, mixed findings) and the reasons for these effects (e.g., transport, product lifetime, user behaviour) were coded to investigate the reasons for the deviation of global warming potentials. Here, the proportion of the reviewed studies that reported potential backfire effects with different reasons were cross analysed with a CE strategy using a heatmap. Finally, the inclusion of life cycle phases in system boundaries (i.e., production, use, end-of-life (EoL), transport, service provision, and infrastructure provision) together with empirical data on these life cycle phases and related parameters (i.e., product lifetime, product substitution, number of users) were obtained from the literature. This information was then used to examine the assumptions and the applications of the empirical data in the existing case studies. Details of data collection items, including codes and levels of analysis, are included in section 1.3 in SI 1.

2.3. Meta-analysis of global warming potentials

A scenario-level analysis (meta-analysis) was conducted to quantitatively analyse the impacts of CE strategies and consumer-oriented product-services on life cycle GHG emissions induced by the introduction of PSS. Typically, a single study typically contains one or more scenarios, i , of different modes of PSS, and reports global warming potentials as absolute impacts (e.g., kgCO₂e) or relative impacts (e.g., use of a standardized indicator) for the base case, E_i^{base} (e.g., conventional sell-out model) and scenario case, E_i^{PSS} (e.g., sharing, rental, or reuse model). In some studies, further data points were presented as a result of sensitivity analysis, e.g., variations due to changes in transport frequency, maintenance type, or product lifetime. All data points (i.e., scenarios and base cases) reported in the reviewed studies were inputted, with the types of data points (i.e., main estimates and sensitivity analysis), types of products, and CE strategies were clearly distinguished from each other. In total, 1513 scenarios (1008 main estimates and 505 sensitivity analyses) were collected for the meta-analysis. In this review, to focus on the most likely scenarios, the meta-analysis comprised mainly of data points reported as main scenarios, whereas data points reported as sensitivity analyses were used to examine the magnitude of backfire effects when they occur.

This review focuses on the expected changes in global warming potentials by introducing CE strategies through PSS. Therefore, the data points for the global warming potentials of scenarios were standardized in the form of improvement rates from the base case $R_i^{improve}$ using the equation

$$R_i^{improve} = \frac{E_i^{PSS} - E_i^{base}}{E_i^{base}} \quad (1)$$

The improvement rates obtained from the literature were examined using descriptive statistics (median, 1st and 3rd quartiles, and the 10th and 90th percentiles) and box plots for different types of product-services and primary CE strategies. Here, if more than two strategies were combined, the analysis was based on the primary strategy which induces the most marked changes in the product use patterns (i.e., following the order as result-, use-, and product-oriented PSS). In addition, to investigate the possibility of backfire effects, the proportion of scenarios with backfire effects, $P_{j,k}^{backfire}$, for CE strategy j and product-service category k were calculated as follows:

$$P_{j,k}^{backfire} = \frac{N_{j,k}^{backfire}}{N_{j,k}^{all}} \quad (2)$$

where, $N_{j,k}^{backfire}$ and $N_{j,k}^{all}$ refer to the number of scenarios with negative

¹ Of the screened articles, one article could not be obtained despite using two document delivery services and was excluded from the analysis.

improvement rates ($R_i^{imp} < 0$) and the number of all reported scenarios for CE strategy j and product-service category k , respectively. Finally, to synthesise the characteristics of the impacts on global warming potentials by different types of PSS, improvement rates were rescaled using the interquartile range (IQR) of the improvement rates for each product-service category k , as follows. Standardisation through centralisation, such as subtracting by the median or mean, was not applied to maintain the influence of positive impacts and backfire effects ($R_i^{imp} < 0$). In the equation

$$R_i^{improve-rescaled} = \frac{R_i^{imp}}{Q_{3,k} - Q_{1,k}} \quad (i \in S_k) \quad (3)$$

S_k refers to scenarios that belong to product-service category k , and $Q_{3,k}$ and $Q_{1,k}$ refer to the 3rd and 1st quartiles of the improvement rates for all scenarios belonging to product-service category k . The descriptive statistics of the rescaled improvement rates and the proportion of scenarios with backfire effects were calculated for all product-service categories, and the results were compared between the CE strategies. More details of the methodology and the coded data set used in the analysis are included in SI 1 and 2, respectively.

2.4. Limitations of the study

As in any study incorporating meta-analysis, this review has limitations, including deviations of assumptions used for the calculations in the reviewed studies, publication bias in the reviewed studies, and the data aggregation method used in the synthesis. The studies reviewed in this paper adopted different system boundaries, data sources, assumptions, impact assessment methods, and geographical and temporal boundaries. Also, the published LCA studies may be affected by publication bias, i.e., they may not reflect the actual range of scenario studies performed. As the purpose of this study was to provide an overview of the characteristics of different types of PSS, rather than provide predictions of quantitative impacts, these variations in parameters were not controlled for or adjusted in the analysis. Moreover, the synthesis analysis in this paper employs the median, 1st and 3rd quartiles, and 10th and 90th percentiles of global warming potential improvement rates and the proportion of scenarios with backfire effects as indicators, and several types of product-services are aggregated into product-service categories that are synthesized through rescaled scores, which clearly omit certain variations within the product-service category and impact categories other than climate change. Furthermore, perspectives other than environmental sustainability, such as acceptability by consumers and business models, were excluded from the scope of this analysis; these other categories may be suitable for further examination in a future study. Nevertheless, the synthesis analysis in this paper provides overview of the extent of potential impacts and current assessment methods employed in PSS.

3. Results

This section provides a systematic review of the literature and the scenarios. First, the methodological approaches of the reviewed studies, including functional units, indicators, and applications of the scenario and sensitivity analyses are presented, followed by a cross analysis of the types of product-services and CE strategies. Then, the results of the meta-analysis of the impacts of PSS on global warming potentials are examined, focusing on potential improvements and backfire effects. Finally, the uses of empirical data and system boundaries are examined.

3.1. Methodological approaches employed by the reviewed studies

3.1.1. Methodologies and objectives

Most of the reviewed studies used LCA as a methodology (LCA 83%, simplified or screening LCA 12%), with a few exceptions (6%) including

input-output LCA and simulations that were not explicitly labelled as being LCA (Fig. 1(a)). Whereas the majority of the studies focussed on comparing different types of PSS and base cases (80%), some studies attempted to quantify the consequences of introducing PSS (13%) (Fig. 1(b)), such as by modelling shifts in the mode of transport after the introduction of a vehicle sharing system [34,35], or modelling imperfect substitutions by purchasing second-hand products [36]. Other studies (16%) focussed on the break-even points in environmental impacts [37–39], optimisation of the PSS [40–42], or identification of key parameters [43].

Among all of the reviewed studies, 35% combined quantification of life cycle impacts with other methods. Some studies adopted methods for collecting empirical data through surveys [36,41,43–54] and/or interviews [45,51,55] (22%) for modelling the flows and stocks of products or materials through material flow analysis (MFA) [41,56–59] (5%), or for employing additional quantitative methods to interpret LCA results (6%), such as through eco-efficiency analyses [60,61] and additional statistical methods [43,62,63]. Other combined methods were performed to design and evaluate PSS alternatives through business model and other frameworks [64–66] (3%) or for analysing consumer behaviours through consumer decision-making models [36,45,56,60] (4%). Whereas the majority of the reviewed studies focussed on case studies of particular systems at a point in time, some studies (11%) extrapolated the results to certain geographical boundaries, such as countries [67,68], regions [69] or worldwide [70], and several studies (6%) conducted dynamic modelling from the present to the future, such as until 2020–2030 [57] and 2035 [71], 2050 [58,65] or 2100 [72].

3.1.2. Functional units and indicators

The functional units used in the reviewed studies could be separated into three types: needs (31%), use (32%), and products (37%) (Fig. 1(c)). Studies using functional units based on needs assume typical needs per person or household for one to a few years related to specific products-services, such as clothing [50], washing [44], driving [64], or drinking [73], and use these as functional units. In some cases, PSS was examined at the aggregated level, and thus the functional units were set as the needs of multiple households [74], target segments [75], or residents in a particular area [76]. Similarly, studies using functional units based on use calculated life cycle impacts for a typical use occasion or multiple use cycles, such as passenger km of mobility [77], the single use of a garment [78], the rental cycle of tools [79], cooking of food [80], amount of water delivered [68], or the provision of media content [49,81,82]. Conversely, studies that used functional units based on product lifetime assumed the typical lifetime of a product, and product life cycle impacts were calculated for the total assumed years of product use. Only 4% of reviewed studies compared results using differing functional units, either as scenarios (e.g., one average use vs. users' needs of clothing [50], passenger km vs. vehicle km [83]), or in the sensitivity analysis (e.g., one-time read vs. screen size of e-reader [49], or product size vs. weight [84]).

The most widely used impact indicators were global warming potentials (81%), which is also the focus of the quantitative synthesis component of this paper (see section 3.3), followed by other environmental impacts (43%), resource depletion and land use (31%), and cumulative energy demand (CED) (27%) (Fig. 1(d)). Use of end-point-level integrated indicators (23%), such as ReCiPe endpoints [85] and Eco-indicators [86], was less prominent in the reviewed literature than individual environmental and resource indicators. Furthermore, the use of economic and social indicators, such as life cycle costs (18%) and job creation (3%) was limited. Several studies that employed MFA used stock indicators, such as number of devices with different use phases, hibernation, or efficiency classes [57,58]; however, beyond these simple stock indicators, there was no implementation of product-level circularity indicators in the reviewed studies.

3.1.3. Scenario and sensitivity analysis

The majority of reviewed studies presented two or more scenarios (74%), with a few exceptions which focused on a single scenario (6%)

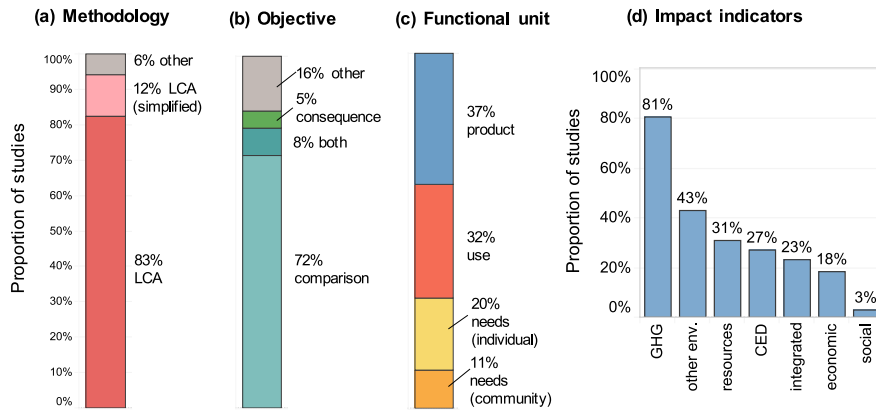


Fig. 1. Methodological approaches adopted by the reviewed studies. (a), (b) and (c) Stacked bar charts on the left indicate the proportion of reviewed studies with relevant methodology, objective, or functional units ($N = 103$). (a) The 'Other' category in the Methodology graph includes simulations, input-output LCA and other methods. (b) 'Both' in Objective refers to studies with objectives involving comparisons and consequences. (c) Needs (individual) and needs (community) refer to studies that used functional units at the individual/household-level and multiple households/community level, respectively. (d) The bar chart at right shows the proportion of reviewed studies using relevant impact indicators, with multiple answers permitted in coding ($N = 103$). Other env.: environmental impacts other than GHG; Resources: resource depletion and land use; CED: Cumulative Energy Demand; Integrated: integrated indicators across multiple impact categories.

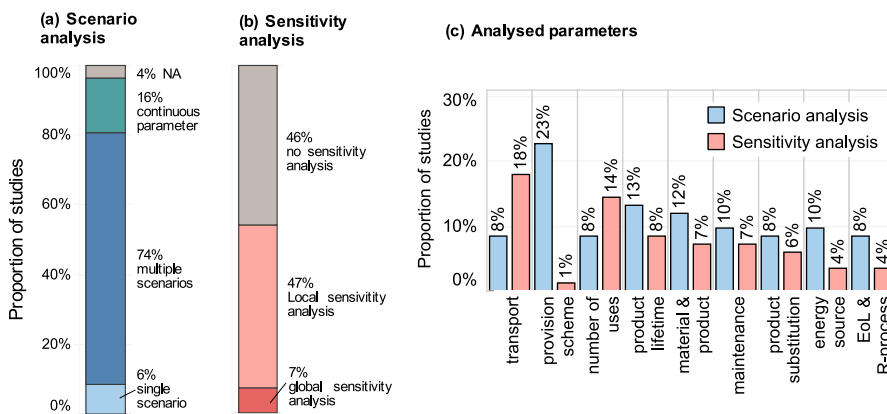


Fig. 2. Scenario and sensitivity analysis of the reviewed studies. Studies using global warming potentials as indicators only ($N = 83$). The stacked bar charts at left, (a) and (b), show the proportion of reviewed studies on the types of scenario or sensitivity analysis. In (a), not applicable (NA) refers to studies that are not relevant as they do not report scenarios, but rather, analyse data for optimisation or identification of key parameters. Continuous parameters refer to studies reporting results that change one or more continuous parameters. (c) The bar chart at right shows the proportion of reviewed studies using relevant parameters in scenario or sensitivity analysis. R-process refers to parameters related to repair, remanufacturing, and refurbishing processes.

(Fig. 2(a)). Apart from these single- and multiple-scenario studies, several studies (16%) investigated changes in life cycle impacts by varying continuous parameters, such as product lifespan [47,87,88], number of uses and demand [49,68,70,89,90], or product substitution rate [91]. More than half of the reviewed studies (54%) conducted sensitivity analyses, especially local sensitivity analyses, by changing individual parameters at the same time (Fig. 2(b)). Only 7% of the reviewed studies conducted global sensitivity analysis to explore differences by systematically varying multiple parameters, typically through Monte Carlo simulation [34,81,92,93]. The parameters examined in the scenario and sensitivity analyses included transport mode and distance; variations in the provision schemes; number and frequency of uses; product lifetime; types of material and products; nature and frequency of maintenance; assumptions related to product substitution; energy sources and efficiency; end-of-life (EoL), repair, remanufacturing, and refurbishing processes; and use-phase consumer behaviours (Fig. 2(c)).

3.2. Circular economy strategies and target product-service categories

3.2.1. Combination of CE strategies and product-service categories

Fig. 3(a) summarises the product-service categories, CE strategies, and PSS types analysed in the reviewed studies (percentages shown at nodes) and their relationships (width of edges). Of the studies analysed, more research has been conducted on product-oriented PSS (67% of reviewed studies) than use- (32%) and result-oriented PSS (16%). Among the product-oriented PSS, reuse was the most studied CE strategy (32%), followed by refurbishing and remanufacturing (19%), and durability (14%); studies on repairing (7%) and upgrading and modularity (6%) were somewhat limited. Studies on product-oriented systems focussed mostly

on energy-using products, such as information and communication technology (ICT) equipment, large appliances, small and medium-sized appliances (59% of product-oriented cases); studies on non-energy-using products, such as clothes and books (25%), were less common.

In use-oriented PSS, rental was the most studied type of CE strategy (24%), followed by pooling (10%), leasing (7%), and C2C sharing (4%). The target products of these studies were somewhat unevenly distributed, with slightly more than half (53% of use-oriented cases) being in the mobility sector, such as automobile carsharing and ridesharing [64, 71,77,91,93–95], and bicycle and scooter sharing² [34,35,48,88,93,96, 97]. In addition, some case studies (30%) were published on energy-using products, such as leasing of appliances [55,65,80], rental of ICT equipment [75], rental and C2C sharing of gardening and do-it-yourself tools [79,98]. Studies on non-energy-using products (12%) were also somewhat limited, e.g., rental of clothes [50,78,99] and books and digital media [82,95]. In addition, some studies focussed on C2C sharing of miscellaneous products, including a variety of appliances, furniture, and tools [54,76].

Case studies on result-oriented PSS were particularly scarce (16%). Examples of such studies include clothes washing services, such as coin laundries [44,45,74], pay-per-use of home appliances [52,65], reading e-books [49,95], online distribution of multimedia contents [81], pay-per-use of cloth diapers [100], and public water fountains [68,73].

As shown in Fig. 3(b), these case studies were somewhat concentrated in Europe (40%), followed by Asia (16%), North America (14%)

² Ridesharing was classified as pooling, whereas car and bicycle/scooter sharing were classified as renting in this review study.

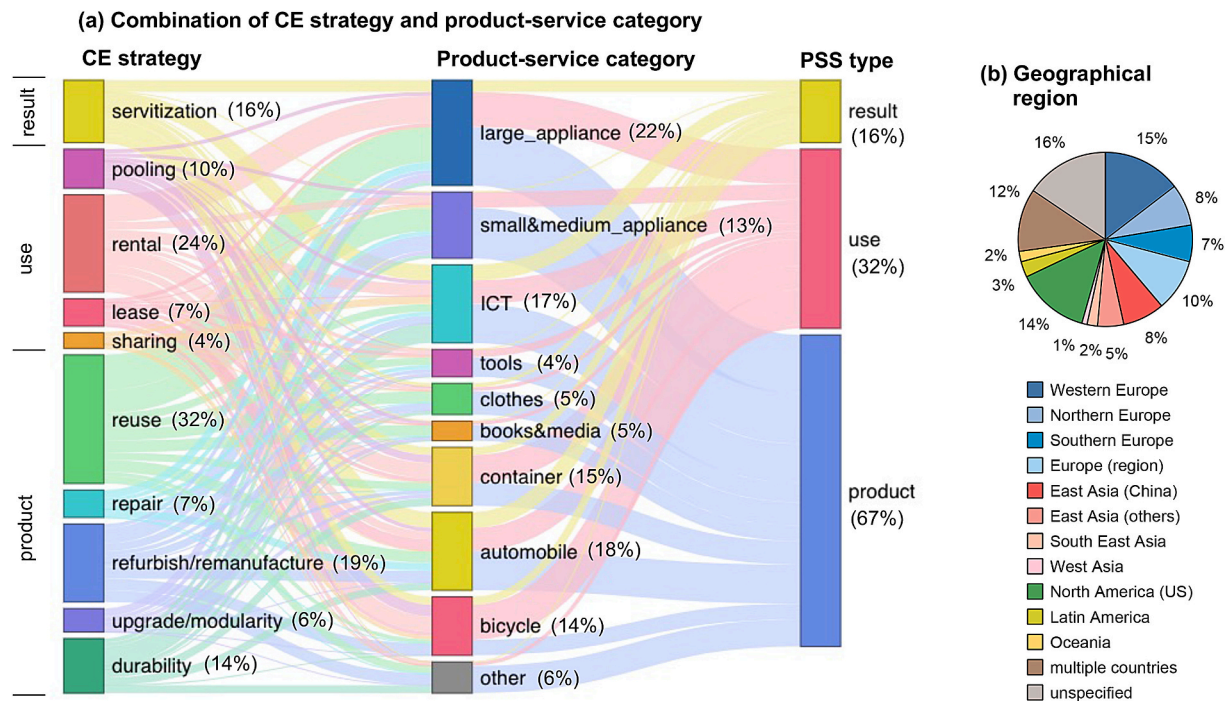


Fig. 3. Target products and circular economy strategies in reviewed studies (a) The percentages indicated at the nodes of the parallel categories diagram represent the proportion of reviewed studies analysing the relevant circular economy (CE) strategy, product-service category, or product service system (PSS) type ($N = 103$). As multiple answers are permitted in coding, the sum of the percentages exceeds 100%. The width of the edges shows the combination of CE strategies/PSS types and product-service categories analysed in the reviewed studies. (b) The pie chart at right shows geographical boundaries of the reviewed studies, not the location of authors.

and other regions (5%). Several studies investigated general trends in PSS impacts without specifying a particular country or region (16%), or studies that targeted two or more countries for comparison [44,68,77, 78,83,93] (12%).

3.2.2. Comparison and integration of multiple CE strategies

As shown in Fig. 4(a), only a limited number of studies (25%) assessed more than one type of CE strategy. These studies compared PSS with different CE strategies (19%), and/or analysed the integration of

multiple strategies in the same system (8%). In particular, studies investigating more than two strategies were rare (9%). Fig. 4(b) summarises the combination of two strategies considered in the literature. Here, the width of the grey- and red-coloured edges represent the number of studies investigating the comparison and integration of strategies shown as nodes, respectively.

The comparative studies formed three clusters: use- and result-oriented PSS (nodes connected by grey edges in the upper-right area around rental, pooling, lease, and servitisation), product-oriented PSS (lower-left area around reuse, refurbishing, and repair), and servitisation of products (horizontal edge between reuse and servitisation). The first cluster was composed of studies that typically compared various types of use- and result-oriented PSS, such as renting, leasing, pooling, and C2C sharing. Examples include studies comparing rental and C2C sharing systems for gardening/do-it-yourself tools [98], pooling and renting automobiles [71,94], and various types of use-oriented strategies of automobiles and books [95]. In some cases, result-oriented PSS was compared with use-oriented PSS, such as downloading and renting of books [95]. In the second cluster, differences within product-oriented systems were examined, such as by comparing upgrading and reusing computers [101]; repairing and upgrading induction cookers [60]; and remanufacturing and refilling ink cartridges [102]. The third cluster compared the impacts of product servitisation with product-oriented PSS. Typical examples included pay-per-use services and reuse models, such as the provision of drinking water [73], reading contents [49], cloth diapers [100]; and pay-per-use and refurbishing of washing machines [52]. In addition, examples of studies comparing more than two strategies were conducted on automobiles (e.g., carpooling, ride-hailing, car sharing, leasing, and taxi [64,93,95]) and books (e.g., reusing, leasing, renting, and C2C sharing [95]).

In addition to these comparative studies, some studies focussed on the integration of multiple CE strategies into the same system. Such studies typically combine product- and use-oriented strategies (red-coloured edges). Examples include rental systems that integrate reuse (e.

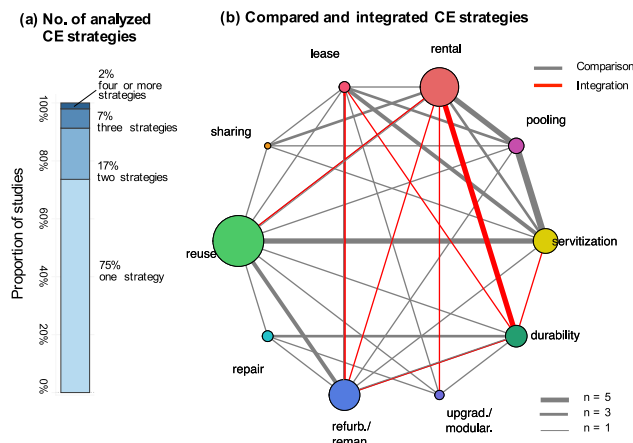


Fig. 4. Comparison and integration of CE strategies. (a) The stacked bar chart at left indicates the number of analysed circular economy (CE) strategies (comparison and integration) for the reviewed literature ($N = 103$). (b) The edges of the network graph shown at right represents the set of the two strategies compared (grey) and integrated (red) in the reviewed literature, with the size of the nodes representing the number of studies. Refurb.: refurbishing; Reman.: remanufacturing; Upgrad.: upgrading; Modular.: modularity.

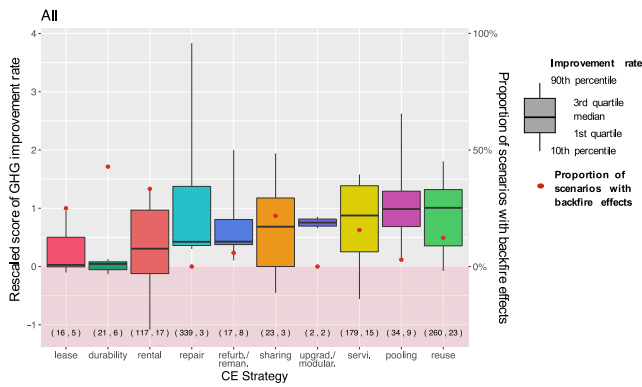


Fig. 5. Rescaled impacts on global warming potentials by adopting circular economy strategies (all products). Main estimates of all product-service categories combined. Rescaled score of improvement refers to the improvement rates rescaled by the interquartile range (IQR) for each product type while maintaining the influence of backfire effects ($N = 1008$). The upper, middle, and lower ranges of boxes refer to 3rd quartile, median, and 1st quartile of the rescaled scores. The upper and lower whiskers indicate the 90th and 10th percentiles of the rescaled scores. The pink-coloured area with rescaled scores lower than 0 refers to backfired scenarios. The red-coloured dots show the proportion of scenarios with backfire effects. The first and second numbers in parenthesis refer to the number of scenarios and studies included in each subcategory, respectively. The ranges between the 1st and 3rd quartiles and the 10th and 90th percentiles indicates the rescaled impacts where 50% and 80% of reported scenarios are positioned. Refurb.: refurbishing; Reman.: remanufacturing; Upgrad.: upgrading; Modular.: modularity; Servi.: servitisation.

g., a cascading rental system of second-hand computers [75]) and refurbished products (e.g., gardening tools [79]); an appliance lease system that was integrated with remanufacturing [103,104]; and the introduction of durable products in bicycle, scooter, and clothes rental

systems [99,105,106].

3.3. Impacts of PSS on global warming potential

This subsection summarises the results of the meta-analysis of the impacts of introducing PSS on global warming potentials. Fig. 5 shows a comparison of rescaled improvement rates of global warming potentials (box plots) and the proportion of scenarios with backfire effects (red-coloured plots) between 10 types of CE strategies as a synthesis of all product-service categories analysed. The improvement rates (percentage of improvements in global warming potentials from baseline) for each product-service category before rescaling are indicated for selected categories in Fig. 6 (see Fig. S1 and Table S3–5 in SI 1 for additional results).

Results-oriented PSS (servitisation) tended to show high potential improvements (median rescaled score: 0.9), but also potentially serious backfire effects when they occurred (10th percentile rescaled score: -0.6) with moderate possibility of backfire effects (proportion of scenarios with backfire effects: 0.2) (Fig. 5). Among non-energy-using products, servitisation showed the highest level of improvement potentials (e.g., $>60\%$ median improvement rates for books and media in Fig. 6(e)). Conversely, for automobiles and large appliances, servitisation (e.g., pay-per-use through ride-hailing and coin laundries) showed only limited improvement potentials (10% or lower median improvement rates) with a higher proportion of backfire effects (Fig. 6(a)(d)).

Among use-oriented PSS, pooling and C2C sharing tended to have moderate to high improvement potentials (median rescaled score: $0.7\text{--}1.0$). However, the risk of backfire effects for these two strategies were different. Whereas pooling had a small risk of backfire effects (proportion of backfire scenarios: 0.03), sharing can potentially cause serious backfire (10th percentile rescaled score: -0.5) with a moderate possibility of backfire effects (proportion of backfire scenarios: 0.2) (Fig. 5). Pooling had the highest improvement potentials among CE strategies for automobiles (approx. 25% median improvement rates) (Fig. 6(d)). Similarly, C2C

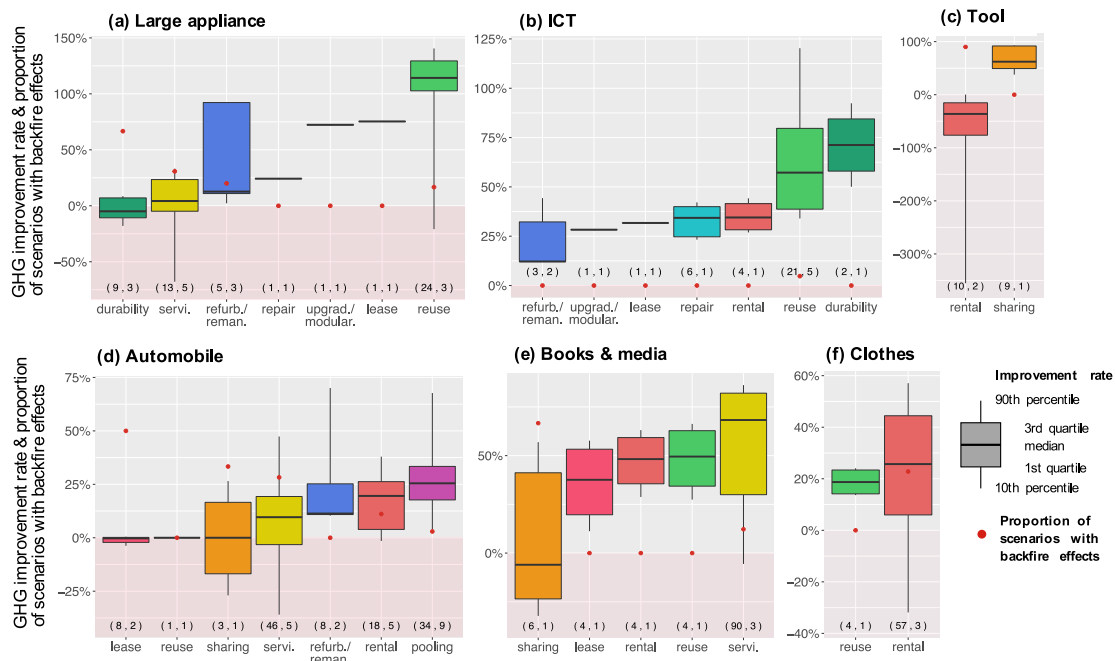


Fig. 6. Impacts on global warming potentials by adopting circular economy strategies (per product-service category) Main estimates only. The upper, middle, and lower ranges of boxes refer to 3rd quartile, median, and 1st quartile of improvement rates ($N = 398$). The upper and lower end of whiskers indicate 90th and 10th percentiles of improvement rates. The pink-coloured area with improvement rates lower than 0% refer to backfired scenarios. The red coloured dots show the proportion of scenarios with backfire effects. The first and second numbers in parenthesis refer to the number of scenarios and studies included in each subcategory, respectively. The ranges between 1st and 3rd quarters and 10th and 90th percentiles represent the improvement rates where 50% and 80% of reported scenarios are positioned. Refurb.: refurbishing; Reman.: remanufacturing; Upgrad.: upgrading; Modular.: modularity; Servi.: servitisation.

sharing showed a high potential for community-based sharing schemes for gardening tools (median improvement rates >60%) (Fig. 6(c)). However, such positive impacts are not always the case; for example, studies on C2C sharing of automobiles and books reported negligible to negative potentials (median improvement rates: approx. 0 to −5%) with a higher risk of backfire effects (Fig. 6(d)(e)).

While the impacts of use-oriented PSS were relatively positive, this was not the case for renting, which had the highest risk of backfire effects among all CE strategies (rescaled score for 10th percentile: −1.1) with relatively high possibility of backfire effects (proportion of backfire scenarios: 0.3) (Fig. 5). Examples of such severe backfire effects are gardening tools, where backfire scenarios account for the majority of reported scenarios (Fig. 6(c)). For non-energy-using products, such as books and clothes, renting showed moderate improvement potentials (median improvement rates: 25–50%), but case studies on rental clothes reported a relatively high risk of backfire effects (Fig. 6(e)(f)). Conversely, leasing was not associated with such a severe possibility of backfire effects (rescaled score for 10th percentile: −0.1), but expected improvements were also limited (median rescaled score: 0.03) (Fig. 5). Leasing appliances, ICT equipment, and books showed small to moderate improvement potentials of 30–75% (Fig. 6(a)(b)(e)); however, leasing automobiles was an exception, having almost negligible impacts on global warming potentials (Fig. 6(d)).

Some product-oriented PSS, such as repairing, upgrading, refurbishing, and remanufacturing showed more reliable improvement potentials with a lower risk of backfire effects. Refurbishing and remanufacturing showed limited, but positive impacts (median rescaled score: 0.4) with a small backfire risk (proportion of backfire scenarios: 0.06) (Fig. 5). The reviewed studies reported small, but positive, improvements for refurbishing and remanufacturing of automobiles, large appliances, and ICT (median improvement rates: approx. 10%) (Fig. 6(a)(b)(d)). Repairing, upgrading and modularity showed relatively high improvement potentials (median rescaled score: 0.4–0.8) without any backfire scenarios reported in the studies reviewed (Fig. 5). This trend was consistent among appliances and ICT (median improvement rates: 25–70%) (Fig. 6(a)(b)).

Such lower risks of backfire effects were not the case for two other product-oriented PSS. Durability was the least impactful strategy (median rescaled score: 0.04, proportion of backfire scenarios: 0.4) (Fig. 5), but marked differences were observed among product-service categories. Although the durability of energy-intensive products (e.g., large appliances) had higher backfire risks, it appears to be an effective strategy for ICT (Fig. 6(a)(b)). Reusing tends to be associated with higher improvement potentials (median rescaled score: 1.0), with a moderate risk of backfire effects (proportion of backfire scenarios: 0.1) (Fig. 5). For non-energy-using products such as books and clothes, reusing can be a positive solution (median improvement potential: 20–50%) (Fig. 6(e)(f)). Reusing energy-using products tends to be associated with high potentials (median improvement potentials for large appliances and ICT: 60–120%); however, the risks of backfire effects are also higher for appliances (Fig. 6(a)(b)).

The above results are a synthesis of the main scenarios and focus on the most likely scenarios investigated in the studies. The results based on all data points, including the data points reported in the sensitivity analysis, are included in Fig. S2–3 and Table S6–8 in SI 1. When the sensitivity analyses are included, some of the potentially negative impacts become more severe (e.g., servitisation, sharing, reusing, renting), implying that the risk of backfire effects could be more serious if more parameter deviations are considered. Nevertheless, the overall conclusions concerning differences in the order of the effects and characteristics of the potentials and backfire risks among the examined strategies are consistent, even when considering the additional data points.

3.4. Reasons for deviations in the global warming potentials and backfire effects

Fig. 7 summarises the major reasons for deviations in the global

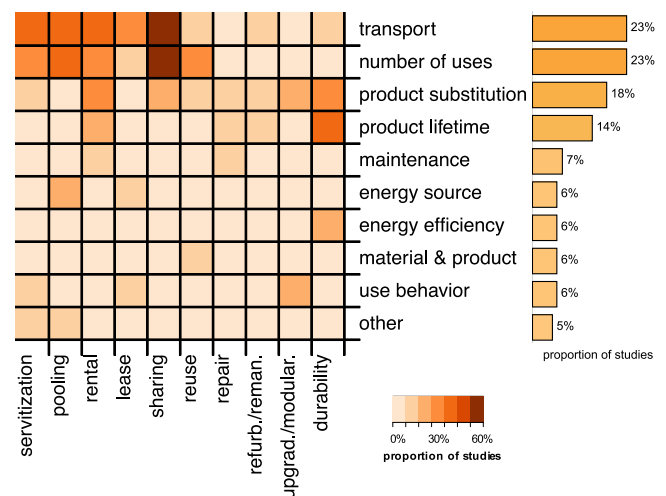


Fig. 7. Reasons for deviations in global warming potentials and backfire effects. Reported reasons of deviations of global warming potentials and potential backfire effects among studies using global warming potentials as indicators ($N = 83$). The width of the bar chart at right shows proportion of studies that reported the reasons corresponding to the rows. The colours in the heatmap at left indicate the proportion of studies that reported the relevant reasons (rows) among the studies analysing each CE strategy (columns). Number of studies reporting no backfire effects or reasons that are included in the denominator to calculate the proportion of studies. Refurb.: refurbishing; Reman.: remanufacturing; Upgrad.: upgrading; Modular.: modularity.

warming potentials reported in the reviewed studies. Overall, transport-related factors (23% of studies) and number of uses (23%) were the most frequently reported reasons for potential backfire effects. These factors were particularly important for use- and result-oriented PSS, including servitisation, pooling, renting, leasing, and C2C sharing (darker cells in upper-left areas of the heatmap shown in Fig. 7). Transport can potentially cause backfire effects, depending on factors, such as transport mode, distance, and efficiency for accessing service spots [73,74] and delivery or pick-up of products [50,79,80,98,99]. A study on C2C sharing also showed the importance of low-carbon logistics for local C2C sharing, such as bike cargo delivery and locker system services [76]. Transport is a key factor for shared mobility, due to rebalancing [34,35,93,107] and additional travel distance [83,91,93]. Similarly, use-related factors (frequency of uses, number of users, and use intensity) were all reported as determinants when renting clothes [50,78] and bicycles [35], and sharing gardening tools [98], vehicle rides [91,95], and miscellaneous products [76]. Apart from use-oriented PSS, the frequency of uses was also a key factor when reusing containers [37,61,67,70,89,90,108] and books [49].

Product substitution was an important factor for almost all types of CE strategies and the third most frequently reported reason for potential backfire effects (18%) (i.e., third row in the heatmap shown in Fig. 7). Here, the reviewed studies reported that the assumptions regarding the type and amount of products replaced due to the introduction of PSS were a crucial factor for various product-services, especially for servitising clothes washing [44,45]; sharing durable products [76]; renting tools [79], clothes [50], and bicycles and scooters [34,35,96,97,106]; and reusing clothes [41,46] and smartphones [36,109]. Product lifetime was the fourth most important reason for potential backfire effects (14%), and was mainly related to product-oriented PSS, such as repairing, refurbishing, durability, and also to renting schemes. Examples of studies affected by this factor include expected product lifetime when renting scooters and bicycles [34,38,88,107], clothes [99], and tools [79] and product age at the time of repairing [47], reusing [87], and refurbishing [110]. In addition, the frequency and types of maintenance (7%) were reported to cause a deviation in global warming

potentials when renting tools [79], bicycles and scooters [34] and cleaning reusable cups [61,70].

Electricity mix and electrification (6%) may influence the performance of PSS, as reported in studies on refurbishing [110], reuse [111], lifetime extension [112], and shared mobility [64,72]. In addition, the energy efficiency of products (6%) was another key reason for potential backfire effects, as illustrated by the case studies on reuse [92] and extending the lifetime [57,112] of appliances. Other reasons for potential backfire effects include user behaviour in the use phase (6%) (e.g., energy and water saving habits when using appliances [44,55]); types of materials and products (6%) (e.g., material choices of containers [37,51,108]; additional electronics for e-readers and e-bikes [49,96]).

3.5. Empirical data and system boundaries

The use of empirical data was limited for multiple lifecycle phases in the reviewed studies (Fig. 8). Compared to the production phase, in which 85% of studies use primary or secondary data, the evaluation of use, transport, and EoL phases were heavily dependent on assumptions (28–32%), so was service provision (12%) whenever it was included in the system boundary; however, as summarised in the previous subsection, transport and use patterns were major reasons for potential backfire effects. Despite number of studies being limited, some studies showed promising approaches for utilising empirical data for key parameters. For example, transport-related empirical data were collected for reusing clothes [46], renting tools [79], leasing appliances [55], and sharing scooters [88,93] from business operation data. Other studies utilised empirical data to estimate use patterns, such as laundry habits [44,45], wearing frequency of rented clothes [50], use patterns of rented tools [79], and use of e-book readers [49] through consumer surveys.

Furthermore, the provision of infrastructure and services were frequently omitted from system boundaries during evaluation, although maintenance was one of the reasons for potential backfire effects reported in the previous subsection. Infrastructure provision was included only in 12% of studies (Fig. 8). Although the number of studies was limited, some studies quantified the impacts of locker systems on product sharing or renting [76,79], docks and stations for bike sharing [35,93,96], water supply infrastructure for public water fountains [73]. Similarly, service provision was included in only half of the reviewed studies (Fig. 8). Various types of services are quantified in the studies, including cleaning services (e.g., for renting clothes [50,78,99], leasing appliances [55], reusable takeaway cups [61,70]), maintenance services (e.g., for sharing scooters and bikes [34,35,88,96,97], sharing automobiles [95]), preparation for reuse (e.g., clothes [46], small and medium appliances [92]), repairing and remanufacturing (e.g., appliances [47,110]), and data consumption (e.g., using ICT products [109], multimedia contents [81], operating rental or sharing platforms [76,79]).

Other parameters related to product-services depended heavily on

assumptions or were not considered in the analysis, although these factors were identified as key reasons of potential backfire effects as summarised in subsection 3.4. Approximately half of the studies rely on assumptions of product lifetime (Fig. 8). Similarly, 26% of the studies rely on assumptions of the number of users, whereas only 15% used primary or secondary data. Examples of using empirical data to estimate the number of users include assessments of the utilisation rate of bicycles [35] and rental tools [79] through collecting data from actual operations or pilot experiments. Moreover, as many as 63% of the reviewed studies did not consider imperfect substitution and focussed on comparisons based on perfect substitution. Even when imperfect substitution is considered, 18% of studies relied on assumptions to estimate the substitution. Although the use of primary data is limited to 12% of studies, some studies employed consumer surveys to estimate the substituted product-services when purchasing second hand products (e.g., smartphones [36], clothes [46]), using rental services (e.g., clothes [50]), using shared mobility (e.g., e-scooters [34,88]), or shifting to pay-per-use schemes (e.g., coin laundries [44,45]).

4. Discussion

4.1. Towards better assessment of PSS from the life cycle perspective

4.1.1. Assessing multiple circular economy strategies for various product-services

The qualitative synthesis of methodological approaches revealed important areas of improvements in the assessment of PSS. Despite previous studies anticipating the largest potentials in reducing environmental impacts from result-oriented PSS [7], the assessment of the life cycle impacts of result-oriented PSS in the consumer product sector has been limited. Only 16% of the studies focussed on result-oriented PSS, and of those, most examined only a limited range of services, such as washing clothes, e-reading, and car driving. Although it is possible that the actual application of result-oriented PSS in society is limited, or not all types of consumer products can be servitised, this lack of research stresses the need for further assessments and pilot implementation of results-oriented PSS. Conversely, the distribution of studies on use-oriented PSS appeared to differ among the types of product-services. For example, while the number of studies on the mobility sector has increased, such as shared vehicles and bicycles, applications in other durable products, such as home appliances have been limited. This limited existence of studies is in sharp contrast to the expected increase in use-oriented PSS in actual society, considering the recent boom in the sharing economy [113].

Importantly, most of the reviewed studies focussed on a single CE strategy, and only 19% of studies compared two or more strategies. Multiple strategy comparisons have been performed in some shared mobility studies [64,93], and another study proposed a comparative

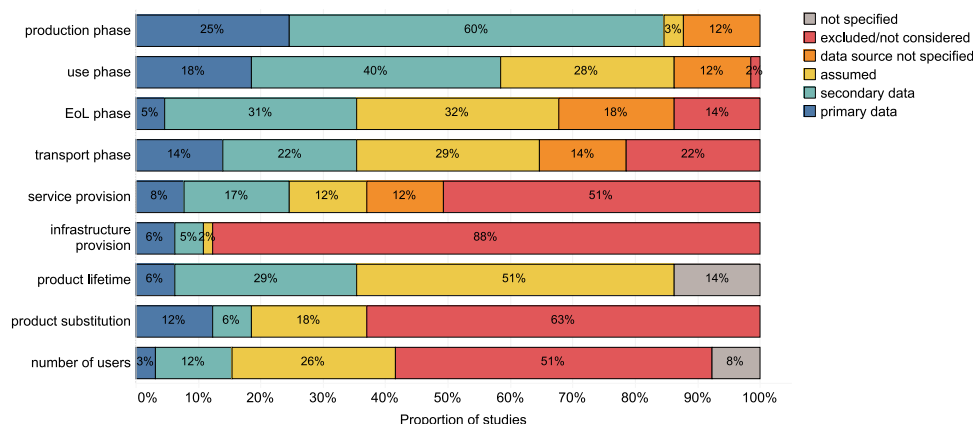


Fig. 8. Uses of empirical data, system boundaries, and related parameters. Studies focussed on comparable global warming potentials only (N = 65). Primary data and secondary data refer to studies obtaining empirical data from primary data collection and from secondary data sources, respectively. ‘Assumed’ refers to studies setting parameters explicitly based on assumptions. ‘Data source not specified’ refers to studies that include relevant life cycle phases as part of system boundary, but the data source is not specified. ‘Excluded/not considered’ refers to studies that exclude relevant life cycle phases from system boundary or do not consider relevant parameters. ‘Not specified’ refers to studies without specifying the details of relevant parameters.

framework across various types of PSS [95]. In addition, given that a PSS is a system designed to satisfy consumer needs efficiently, one PSS may not necessarily be limited to only one CE strategy. Instead, integration of multiple CE strategies into one PSS could potentially improve the performance and efficiency of PSS. For example, in this review study, product lifetime was reported to be one of the key reasons for potential backfire effects, which can be controlled by adding another strategy, such as reusing, upgrading, and refurbishing. However, only 8% of the studies incorporated this approach, typically in cases of combining product-oriented strategies into use-oriented PSS, such as a rental system that integrated reuse [75] or a leasing system that integrated remanufacturing [103,104]. Considering the variety of CE strategies that have been discussed in the literature and policymaking [11,12], more studies comparing or integrating multiple CE strategies through PSS should be undertaken in the future.

4.1.2. Assessing the consequences of PSS introduction with reconsideration of functional units

Most of the reviewed studies focussed on comparisons of PSS scenarios with baselines (such as in a traditional sales model) assuming perfect substitution. The application of consequential and other approaches was limited to only 28% of the reviewed studies, and imperfect substitution was only considered in 37% of the studies. This indicates that, in contrast to the three objectives of LCA of PSS proposed in existing guidelines [6,21], the published case studies are heavily skewed toward comparative approaches. Considering that assumptions for product substitution were identified as the third most important determinant of potential backfire effects related to almost all types of PSS, this predominant approach may impose limitations on existing case studies. Some of the reviewed studies performed comparisons using more precise modelling of imperfect substitutions, as illustrated by the fact that 8% of studies focussed on both comparisons and consequences. In this review, some promising approaches for estimating the substituted amount and types of product-services were identified, such as the use of consumer surveys [34,36,44–46,50]. Although there is still no agreement on a standard methodology for modelling imperfect substitution, approaches based on consumer surveys should be further strengthened in future.

Parameters that are often internalised in functional units appeared to affect the results of PSS assessments. Defining functional units has been identified as a major challenge in applying LCA into PSS [32]. Although several approaches have been proposed [6,114,115], there has been no agreement on how to define functional units when evaluating PSS. Through the systematic review, it was found that assumptions behind the functional units tend to overlap with critical parameters. Studies employing functional units based on the total amount of needs per household often assume that the total amounts of physical consumption for certain duration are fixed to the current average or typical levels. Similarly, functional units based on use occasion inexplicitly assume that the total number of uses are not affected by PSS, and that use patterns are fixed to average or typical behaviours. Needless to say, studies using functional units based on product lifetime typically assume a typical or average product lifetime. However, the factors related to these assumptions behind functional units, such as the number of uses and product lifetime, were identified as being among the most important determinants of potential backfire effects. Interestingly, only 4% of the reviewed studies compared or conducted sensitivity analyses on the different selections of functional units. Although there is no unanimous approach among the reviewed studies, it was confirmed that improvements to functional units and scenario development are needed instead of the ones used in conventional product-focused LCA.

4.1.3. Integrating product-level circularity assessments and non-environmental aspects into LCA

The coverage of indicators illustrates the potential for incorporating product-level circularity assessments and aspects other than environmental impacts into PSS assessments. Measuring the progress of CE

based on life cycle-thinking approaches is not limited to environmental impacts; these measurements can also be used to analyse physical properties in technological cycles, such as product-level circularity [116]. While it was expected that the reviewed studies would have relevant information on product-level circularity as part of life cycle inventories, only a few studies reported product-level material flows and stock indicators [57,58]; none of the studies applied more comprehensive, product-level circularity indicators. Some existing indicators, such as Material Circularity Indicator [117], Product-Level Circularity Metric [118], and Total Restored Products [119], can be integrated into PSS evaluations, in addition to impact indicators. Furthermore, the potential impacts of introducing PSS on climate change mitigation and other environmental impacts have been relatively well evaluated, as 81% and 43% of the reviewed studies use global warming potentials and other environmental impacts as indicators, respectively. Conversely, relatively few studies considered costs and job creation as part of the quantitative assessments. Considering that the contribution to the economy and job creation have been emphasised as potential benefits of CE [120], indicators beyond environmental impacts could be utilised more frequently in assessing PSS from a life cycle perspective.

4.1.4. Utilising empirical data for critical modelling parameters with global sensitivity analysis

The various factors determining potential rebound and backfire effects identified in this review study, from background energy systems to user behaviours, would impose challenges in modelling practices. Despite the importance of considering numerous parameters, as many as half of the studies did not conduct sensitivity analysis. Indeed, in cases when sensitivity analysis was conducted, it focussed mainly on local sensitivity analysis by changing only individual parameters at the same time. Global sensitivity analysis examines the contribution of each input parameter on the variance of outputs and considers the probability density and uncertainty propagation [121], making such studies more useful for a more concrete interpretation compared to local sensitivity analysis [122]. As introducing PSS is expected to increase uncertainty due to systemic effects beyond the typical product-sales system, global sensitivity analysis could be better suited to understanding the expected benefits and risks more precisely, and to identify key parameters for optimal PSS design and use.

Such analytical efforts should be supported by empirical data. Indeed, the limited use of empirical data for key parameters and omissions of services and infrastructure from system boundaries when assessing PSS could be problematic in efforts to more accurately model PSS and propose an optimised design for sustainable PSS. This review showed that critical parameters related to potential backfire effects, including transport, number of uses, product lifetime, and product substitution, all rely heavily on assumptions. However, in some cases, since PSS has either only recently been implemented in society, or it does not exist, empirical data collection may be challenging. Nevertheless, some of the reviewed studies showed potential for collecting and utilising actual operation data [46,55,79,88,93] and consumer survey data [44,45,49,50]. In addition, service and infrastructure provision were not considered in most of the reviewed studies, despite the fact that service provision in areas such as maintenance was also identified as a key reason for backfire effects in this study. Recent findings from input-output analysis revealed that the impact of infrastructure provision cannot be ignored, and may increase the environmental impacts and resource consumption in the service sector if properly allocated [123–125]. PSS may change the ways in which product-services are delivered, used, and collected. Consequently, the necessary types and level of infrastructure could be potentially changed. The accumulation of data, partnerships with business sectors, and use of survey methods should be strengthened in future studies in order to better model these key parameters, services, and infrastructures.

4.2. Leveraging circular economy strategies for decarbonisation

4.2.1. Reconsidering the importance of low-risk circular economy strategies

The meta-analysis of global warming potentials identified different orders of impacts and trade-offs between improvement and backfire effects among CE strategies. Overall, CE strategies that showed moderate to high improvement potentials (median rescaled score: 0.4–1.0) with limited backfire risks (proportion of backfire scenarios: up to 0.06) are pooling and some product-oriented strategies, i.e. upgrading, modularity, repairing, refurbishing, and remanufacturing (Fig. 5). Conversely, some use-, product-, and result-oriented strategies, such as servitising, C2C sharing, and reusing, have much higher backfire risks (proportion of backfire scenarios: 0.1–0.2); however, these strategies also showed high improvement potentials (median rescaled score: 0.7–1.0). Compared to these strategies, durability and remaining use-oriented strategies, i.e., renting and leasing, showed limited improvement potentials. While merely extending lifetime (durability) had limited improvement potentials (median rescaled score: 0.04), renting and leasing showed contrasting characteristics; renting had higher improvement potentials and serious backfire risks (median rescaled score: 0.3, rescaled score for 10th percentile: −1.1) and leasing had lower backfire risks and improvement potentials (median rescaled score: 0.03, rescaled score for 10th percentile: −0.1). Considering these trade-offs and order of impacts, some lower-risk strategies with high improvement potentials (e.g., repairing, upgrading, modularity, refurbishing, remanufacturing, and pooling) should be more extensively utilised. This recommendation for prioritising product-oriented PSS is in contrast with previous studies, which proposed that priority should be given to result- and use-oriented strategies [7]; rather, this recommendation is in line with more recent studies which highlighted the risks of rebound effects [18] and recent CE policy developments emphasising various product-oriented strategies [12].

4.2.2. Avoiding backfire effects from high-risk circular economy strategies

This meta-analysis also revealed the challenges in addressing potential rebound and backfire effects in contrast to the expected large-factor theoretical potentials from PSS. In contrast to a previous study, which envisaged radical improvements in environmental impacts (up to a factor of 10) from result-oriented PSS, considerable improvement (up to a factor of two) from use-oriented PSS, and incremental improvement (10–20%) from product-oriented PSS [7], in terms of global warming potentials, most of the reviewed studies typically achieved incremental to considerable reductions, but not radical reductions. Even for the most successful product-service categories, the median improvement rates observed in the reviewed studies were up to around 50–70% (e.g., servitisation of books and media, reusing non-energy-using products, and upgrading and repairing ICT and appliances). Although these limited improvement rates are only among the reviewed studies and do not impose an ultimate limitation on the potential of PSS in the future, the findings suggest that more efforts are needed to decarbonise the life cycle impact of consumer products. Implementing only aforementioned low-risk strategies may not be sufficient for the decarbonisation envisaged by the Paris Agreement, considering the need to significantly reduce household carbon footprints [126–128]. Therefore, in addition to low-risk CE strategies, a well-designed PSS incorporating higher-risk strategies with high improvement potentials (e.g., servitisation, sharing, reusing) should also be promoted in order to achieve further reductions in GHG emissions on the condition that backfire effects are avoided.

Minimizing rebound effects and avoiding backfire effects are crucial when introducing many of the assessed CE strategies. Although previous studies emphasised that PSS could potentially have environmentally negative consequences [5,7], no systematic review has been conducted on the concrete reasons for rebound and backfire effects. The major reasons for the potential backfire effects identified in this study, such as transport, number of uses, product substitution and lifetimes, maintenance, energy source and efficiency, material and products, and use behaviour, should be

incorporated into the assessment models in future studies and reflected into the design of sustainable PSS. Furthermore, some of the identified factors, e.g., energy efficiency and transport modes, are expected to change significantly in the future due to a transition to decarbonisation. Some of the reviewed studies emphasised the importance of this perspective, as illustrated by the need for low-carbon delivery and pick-up systems for local sharing [76], as well as the positive synergies of product reuse and lifetime extension and greener electricity during the use phase [110]. The future implementation of PSS may need to be designed to incorporate decarbonised transport and energy systems, and to control the reasons for potential rebounds.

4.2.3. Integrating multiple circular economy strategies

Further reductions in GHG and rebound effects could be potentially achieved through the integration of multiple CE strategies, such as product- and use-oriented PSS. Such an approach could potentially redouble the sustainability impacts, overcoming the limited potentials resulting from the adoption of a single strategy, as discussed above. If the positive performance observed for repairing, upgrading, modularity, refurbishing, and remanufacturing could be utilised, then the backfire risks of renting and limited improvement potentials of leasing could be potentially addressed, as well as other use-oriented strategies, such as sharing and pooling. This potential synergy between multiple strategies is partly related to rebound effects, such as product lifetime and maintenance, and could be addressed through additional CE strategies. Although assessments involving multiple strategy integration were limited, some of the reviewed studies showed potential improvement through renting refurbished [79] or reused products [75], as well as through increasing the durability of products for renting or leasing [65, 99,105]. The precise impact of integrating multiple strategies could not be evaluated in this meta-analysis due to the limitation of data. However, further application and assessments of multi-strategy integration could potentially improve the performance of PSS.

5. Conclusion

Assessments based on a life cycle approach are central to the design of effective solutions for the development of a circular and decarbonised economy. The importance of the potential nexus of CE and climate change mitigation cannot be emphasised enough, considering the roles of resource-efficient provision systems for decarbonation. Although improvement potentials from specific CE strategies quantified in this study are based on the currently available set of applications and assessments of PSS, the findings revealed that strategic promotion of CE in the consumer product sector could be useful for climate change mitigation. While previous review studies on the sustainability impacts from PSS and its assessment methodology have focused mostly on qualitative analysis, conceptual discussion, and a particular CE strategy or product, this review paper is considered novel for the two major contributions summarised below.

First, the present study identified the major shortcomings in existing case studies on the life cycle impacts from PSS and the points that need to be addressed in future studies based on a systematic review of the 103 case studies. Specifically, weakly reflected perspectives among the case studies were identified, i.e., comparisons of multiple, or a combination of, CE strategies (e.g., rental with reuse, leasing with remanufacturing) was proposed instead of focusing on single strategy, and covering use-oriented PSS in the non-mobility sector and result-oriented PSS. Furthermore, it revealed that assessments of the consequences of introducing PSS, rather than case comparisons, have been emphasised less in the literature, and imperfect substitutions between linear products and newly introduced PSS have been rarely considered. In addition, none of the reviewed studies reported product-level circularity indicators beyond simple material flows and stock indicators, whereas impact categories other than climate change and environmental impacts, such as life cycle costs and job creation, have been examined less frequently. Finally, the collection and use of empirical

data, especially data related to reasons for potential backfire effects, such as transport, number of uses, product lifetime, and product substitutions, are critically lacking. There is thus an opportunity to utilize more sophisticated assessment methods incorporating global sensitivity analysis with additional data collection methods (e.g., consumer surveys and automatically collected data from service providers), expanding system boundaries to consider service and infrastructure provision, and reconsidering the functional units that are closely related to key backfire and rebound parameters.

Second, the present study provided quantitative synthesis of GHG reduction potentials, possibility of backfire effects, and major reasons for backfire effects by introducing PSS based on more than 1500 scenarios identified from the existing case studies. In contrast to the presumption that the result- and use-oriented PSS only have large improvement potentials, some product-oriented strategies (e.g., upgrading, modularity, repairing, refurbishing, and remanufacturing) and pooling showed moderate to high improvement potentials with limited backfire risks. Conversely, some use-, product-, and result-oriented strategies (servitising, sharing, and reusing) showed higher backfire risks, but their improvement potentials could be increased if such backfire effects are controlled. In addition, this study identified key reasons for rebound and backfire effects resulting from the introduction of PSS, such as changes in the number of uses, transport, product substitution and lifetime, maintenance, energy source and efficiency, materials and products, and user behaviour; controlling these parameters is a critical precondition for utilising high-potential strategies. Furthermore, the combination of multiple CE strategies (e.g., combination of use- and product-oriented PSS) could be potentially beneficial, as it could combine the potential impact and address reasons for backfire effects, such as product lifetime. However, the precise impacts from such combinations may need to be assessed further in future studies. The scope of the systematic review conducted in this study is highly comprehensive, covering 10 types of CE strategies (reuse, repair, refurbishing/remanufacturing, upgrading/modularity, durability, pooling, renting, leasing, sharing, and servitisation) applied to various consumer durable and semi-durable products (large appliances, small and medium appliances, ICT equipment, tools, clothes, books/media, containers, automobiles, and bicycles/scooters), which enabled cross-cutting assessments of methodologies and sustainability impacts from consumer-oriented PSS.

These findings stress the need for policymakers and business sectors to reconsider the priorities associated with utilising CE for decarbonisation. For example, it is recommended that low-risk and high-potential strategies be prioritised to achieve higher rates of GHG reduction while avoiding backfire effects and minimizing rebound effects, rather than simply assuming that all types of CE strategies are equally effective. Especially for strategies for which higher risks were identified, further assessments may be needed to avoid backfire effects and minimise rebound effects. The specific CE strategies to be prioritised may be subject to change and depend on the implementation of new PSS, trends in consumer behaviour, business models, technology, and transport and energy systems, reflecting the transition to decarbonisation in the future. Therefore, further accumulation of empirical data, pilot studies, and scenario or simulation analyses with improved assessment approaches that consider the anticipated changes are considered important. Research gaps that can be addressed in future review studies include analysing the effects of combining multiple CE strategies, considering impact categories other than climate change, applying alternative methods for synthesizing quantitative review results, and examining the potential effects of publication biases and differences in methodologies. The areas for methodological improvements proposed in this study, e.g., covering multiple strategies, assessing consequences, integrating circularity assessments, and utilising empirical data, can strengthen the design and assessment of sustainable PSS, which in turn can facilitate implementation of the CE strategies focused on decarbonised production and consumption.

Funding

This work was supported by Grants-in-Aid for Scientific Research from the Japan Society for the Promotion of Science [Grant numbers: JP21K12374 and JP19H04325].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful to Mizuho Owashi for assistance with identifying literature search terms.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rser.2021.111858>.

References

- [1] Vargo SL, Lusch RF. Evolving to a new dominant logic for marketing. *J Mark* 2004;68:1–17. <https://doi.org/10.1509/jmkg.68.1.1.24036>.
- [2] Hossain M. Sharing economy: a comprehensive literature review. *Int J Hosp Manag* 2020;87:102470. <https://doi.org/10.1016/j.ijhm.2020.102470>.
- [3] Barefoot K, Curtis D, Jolliffe W, Nicholson JR, Omohundro R. Defining and measuring the digital economy. Working Paper. Bureau of Economic Analysis, US Department of Commerce; 2018.
- [4] Tukker A, Tischner U. Product-services as a research field: past, present and future. Reflections from a decade of research. *J Clean Prod* 2006;14:1552–6. <https://doi.org/10.1016/j.jclepro.2006.01.022>.
- [5] Mont OK. Clarifying the concept of product-service system. *J Clean Prod* 2002;10:237–45. [https://doi.org/10.1016/S0959-6526\(01\)00039-7](https://doi.org/10.1016/S0959-6526(01)00039-7).
- [6] Kjaer LL, Pigosso DCA, McAloone TC. A guide for evaluating the environmental performance of Product/Service-Systems. Technical University of Denmark; 2017.
- [7] Arnold T. Eight types of product-service system: eight ways to sustainability? Experiences from suspronet. *Bus Strat Env* 2004;13:246–60. <https://doi.org/10.1002/bse.414>.
- [8] Michellini G, Moraes RN, Cunha RN, Costa JMH, Ometto AR. From linear to circular economy: PSS conducting the transition. *Procedia CIRP* 2017;64:2–6. <https://doi.org/10.1016/j.procir.2017.03.012>.
- [9] Khan I, Hou F, Le HP, Ali SA. Do natural resources, urbanization, and value-adding manufacturing affect environmental quality? Evidence from the top ten manufacturing countries. *Resour Pol* 2021;72:102109. <https://doi.org/10.1016/j.resourpol.2021.102109>.
- [10] Ngan SL, How BS, Teng SY, Promentilla MAB, Yatim P, Er AC, et al. Prioritization of sustainability indicators for promoting the circular economy: the case of developing countries. *Renew Sustain Energy Rev* 2019;111:314–31. <https://doi.org/10.1016/j.rser.2019.05.001>.
- [11] Blomsma F, Kjaer L, Pigosso D, McAloone T, Lloyd S. Exploring circular strategy combinations - towards understanding the role of PSS. *Procedia CIRP* 2018;69:752–7. <https://doi.org/10.1016/j.procir.2017.11.129>.
- [12] European Commission. A new circular economy action plan for a cleaner and more competitive Europe. COM(2020) 98 final; 2020.
- [13] Finkbeiner M, Bach V. Life cycle assessment of decarbonization options—towards scientifically robust carbon neutrality. *Int J Life Cycle Assess* 2021;26:635–9. <https://doi.org/10.1007/s11367-021-01902-4>.
- [14] Cantzler J, Creutzig F, Ayargarnchanakul E, Javaid A, Wong L, Haas W. Saving resources and the climate? A systematic review of the circular economy and its mitigation potential. *Environ Res Lett* 2020;15:123001. <https://doi.org/10.1088/1748-9326/abb7>.
- [15] Ellen MacArthur Foundation. Completing the picture: how the circular economy tackles climate change. 2019.
- [16] Ivanova D, Stadler K, Steen-Olsen K, Wood R, Vita G, Tukker A, et al. Environmental impact assessment of household consumption. *J Ind Ecol* 2016;20:526–36. <https://doi.org/10.1111/jiec.12371>.
- [17] Ellen MacArthur Foundation. Towards the Circular Economy: opportunities for the consumer goods sector. 2013.
- [18] Tukker A. Product services for a resource-efficient and circular economy - a review. *J Clean Prod* 2015;97:76–91. <https://doi.org/10.1016/j.jclepro.2013.11.049>.
- [19] Hertwich EG. Consumption and the rebound effect - an industrial ecology perspective. *J Ind Ecol* 2005;9:85–98. <https://doi.org/10.1162/1088198054084635>.

- [20] Winans K, Kendall A, Deng H. The history and current applications of the circular economy concept. *Renew Sustain Energy Rev* 2017;68:825–33. <https://doi.org/10.1016/j.rser.2016.09.123>.
- [21] Kjaer LL, Pigosso DCA, McAloone TC, Birkved M. Guidelines for evaluating the environmental performance of Product/Service-Systems through life cycle assessment. *J Clean Prod* 2018;190:666–78. <https://doi.org/10.1016/j.jclepro.2018.04.108>.
- [22] Chen Z, Huang L. Application review of LCA (life cycle assessment) in circular economy: from the perspective of PSS (product service system). *Procedia CIRP* 2019;83:210–7. <https://doi.org/10.1016/j.procir.2019.04.141>.
- [23] Boehm M, Thomas O. Looking beyond the rim of one's teacup: a multidisciplinary literature review of product-service systems in information systems, business management, and engineering & design. *J Clean Prod* 2013;51:245–60. <https://doi.org/10.1016/j.jclepro.2013.01.019>.
- [24] Piontek FM, Müller M. Literature reviews: life cycle assessment in the context of product-service systems and the textile industry. *Procedia CIRP* 2018;69:758–63. <https://doi.org/10.1016/j.procir.2017.11.131>.
- [25] Lieder M, Rashid A. Towards circular economy implementation: a comprehensive review in context of manufacturing industry. *J Clean Prod* 2016;115:36–51. <https://doi.org/10.1016/j.jclepro.2015.12.042>.
- [26] Khan MA, Mittal S, West S, Wuest T. Review on upgradability e A product lifetime extension strategy in the context of product service systems. *J Clean Prod* 2018;204:1154–68. <https://doi.org/10.1016/j.jclepro.2018.08.329>.
- [27] Zumsteg JM, Cooper JS, Noon MS. Systematic review checklist: a standardized technique for assessing and reporting reviews of life cycle assessment data. *J Ind Ecol* 2012;16:S12–21. <https://doi.org/10.1111/j.1530-9290.2012.00476.x>.
- [28] Rocha MH, Capaz RS, Lora EES, Nogueira LAH, Leme MMV, Renó MLG, et al. Life cycle assessment (LCA) for biofuels in Brazilian conditions: a meta-analysis. *Renew Sustain Energy Rev* 2014;37:435–59. <https://doi.org/10.1016/j.rser.2014.05.036>.
- [29] Minunno R, O'Grady T, Morrison GM, Gruner RL. Investigating the embodied energy and carbon of buildings: a systematic literature review and meta-analysis of life cycle assessments. *Renew Sustain Energy Rev* 2021;143:110935. <https://doi.org/10.1016/j.rser.2021.110935>.
- [30] Cooper DR, Gutowski TG. The environmental impacts of reuse: a review. *J Ind Ecol* 2017;21:38–56. <https://doi.org/10.1111/jiec.12388>.
- [31] Ivanova D, Barrett J, Wiedenhofer D, Macura B, Callaghan M, Creutzig F. Quantifying the potential for climate change mitigation of consumption options. *Environ Res Lett* 2020;15:093001. <https://doi.org/10.1088/1748-9326/ab8589>.
- [32] Kjaer LL, Pagoropoulos A, Schmidt JH, McAloone TC. Challenges when evaluating product/service-systems through life cycle assessment. *J Clean Prod* 2016;120:95–104. <https://doi.org/10.1016/j.jclepro.2016.01.048>.
- [33] Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;6:e1000097. <https://doi.org/10.1371/journal.pmed.1000097>.
- [34] Hollingsworth J, Copeland B, Johnson JX. Are e-scooters polluters? the environmental impacts of shared dockless electric scooters. *Environ Res Lett* 2019;14:084031. <https://doi.org/10.1088/1748-9326/ab2da8>.
- [35] Luo H, Kou Z, Zhao F, Cai H. Comparative life cycle assessment of station-based and dock-less bike sharing systems. *Resour Conserv Recycl* 2019;146:180–9. <https://doi.org/10.1016/j.resconrec.2019.03.003>.
- [36] Makov T, Vivanco DF. Does the circular economy grow the pie? The case of rebound effects from smartphone reuse. *Front Energy Res* 2018;6:39. <https://doi.org/10.3389/fenrg.2018.00039>.
- [37] Gallego-Schmid A, Mendoza JMF, Azapagic A. Environmental impacts of takeaway food containers. *J Clean Prod* 2019;211:417–27. <https://doi.org/10.1016/j.jclepro.2018.11.220>.
- [38] Chen J, Zhou D, Zhao Y, Wu B, Wu T. Life cycle carbon dioxide emissions of bike sharing in China: production, operation, and recycling. *Resour Conserv Recycl* 2020;162:105011. <https://doi.org/10.1016/j.resconrec.2020.105011>.
- [39] Prakash S, Liu R, Schischke K, Stobbe PL. Early replacement of notebooks considering environmental impacts. *Electronics Goes Green* 2012+ (EGG); 2012.
- [40] Solano CM, Roldán RF, Carvajal MF, Gómez AJ, Mattos S, Vives JI. Reverse logistic processes for glass container reuse. *Environ Process* 2021;8:397–411. <https://doi.org/10.1007/s40710-020-00482-z>.
- [41] Fortuna LM, Diyamandoglu V. Optimization of greenhouse gas emissions in second-hand consumer product recovery through reuse platforms. *Waste Manag* 2017;66:178–89. <https://doi.org/10.1016/j.wasman.2017.04.032>.
- [42] Luo H, Zhao F, Chen WQ, Cai H. Optimizing bike sharing systems from the life cycle greenhouse gas emissions perspective. *Transport Res C Emerg Technol* 2020;117:102705. <https://doi.org/10.1016/j.trc.2020.102705>.
- [43] Bovea MD, Ibáñez-Forés V, Juan P, Pérez-Belis V, Braulio-Gonzalo M. Variables that affect the environmental performance of small electrical and electronic equipment. Methodology and case study. *J Clean Prod* 2018;203:1067–84. <https://doi.org/10.1016/j.jclepro.2018.08.240>.
- [44] Moon D, Amasawa E, Hirao M. Consumer motivation and environmental impact of laundry machine-sharing: analysis of surveys in Tokyo and Bangkok. *Sustainability* 2020;12:9756. <https://doi.org/10.3390/su1229756>.
- [45] Amasawa E, Suzuki Y, Moon D, Nakatani J, Sugiyama H, Hirao M. Designing interventions for behavioral shifts toward product sharing: the case of laundry activities in Japan. *Sustainability* 2018;10:2687. <https://doi.org/10.3390/su10082687>.
- [46] Farrant L, Olsen SI, Wangel A. Environmental benefits from reusing clothes. *Int J Life Cycle Assess* 2010;15:726–36. <https://doi.org/10.1007/s11367-010-0197-y>.
- [47] Bovea MD, Ibáñez-Forés V, Pérez-Belis V. Repair vs. replacement: selection of the best end-of-life scenario for small household electric and electronic equipment based on life cycle assessment. *J Environ Manag* 2020;254:109679. <https://doi.org/10.1016/j.jenvman.2019.109679>.
- [48] Zheng F, Gu F, Zhang W, Guo J. Is bicycle sharing an environmental practice? Evidence from a life cycle assessment based on behavioral surveys. *Sustainability* 2019;11:1550. <https://doi.org/10.3390/su11061550>.
- [49] Tahara K, Shimizu H, Nakazawa K, Nakamura H, Yamagishi K. Life-cycle greenhouse gas emissions of e-books vs. paper books: a Japanese case study. *J Clean Prod* 2018;189:59–66. <https://doi.org/10.1016/j.jclepro.2018.03.321>.
- [50] Johnson E, Plepys A. Product-service systems and sustainability: analysing the environmental impacts of rental clothing. *Sustainability* 2021;13:2118. <https://doi.org/10.3390/su13042118>.
- [51] Boesen S, Bey N, Niero M. Environmental sustainability of liquid food packaging: is there a gap between Danish consumers' perception and learnings from life cycle assessment? *J Clean Prod* 2019;210:1193–206. <https://doi.org/10.1016/j.jclepro.2018.11.055>.
- [52] Bressanelli G, Perona M, Saccani N. Benefits and acceptance rate of circular economy alternatives: evidences from a user survey in the washing machine industry. *Proc XXIV Summer School Francesco Turco* 2019:245–52.
- [53] Rosenthal C, Fatimah YA, Biswas WK. Application of 6R principles in sustainable supply chain design of western Australian white goods. *Procedia CIRP* 2016;40:318–23. <https://doi.org/10.1016/j.procir.2016.01.048>.
- [54] Khajehzadeh I, Vale B. The share of furniture and appliances in the life cycle energy of New Zealand houses. *Proc 33rd PLEA Int Conf* 2017;3:3555–62.
- [55] Chun YY, Lee KM. Environmental impacts of the rental business model compared to the conventional business model: a Korean case of water purifier for home use. *Int J Life Cycle Assess* 2017;22:1096–108. <https://doi.org/10.1007/s11367-016-1227-1>.
- [56] Streicher-Porte M, Marthaler C, Böni H, Schluep M, Camacho Á, Hilty LM. One laptop per child, local refurbishment or overseas donations? Sustainability assessment of computer supply scenarios for schools in Colombia. *J Environ Manag* 2009;90:3498–511. <https://doi.org/10.1016/j.jenvman.2009.06.002>.
- [57] Glöser-Chahoud S, Pfaff M, Schultmann F. The link between product service lifetime and GHG emissions: a comparative study for different consumer products. *J Ind Ecol* 2021;25:465–78. <https://doi.org/10.1111/jiec.13123>.
- [58] Boldoczki S, Thorenz A, Tuma A. Does increased circularity lead to environmental sustainability?: the case of washing machine reuse in Germany. *J Ind Ecol* 2021;25:864–76. <https://doi.org/10.1111/jiec.13104>.
- [59] Bobba S, Tecchio P, Ardente F, Mathieux F, dos Santos FM, Pekar F. Analysing the contribution of automotive remanufacturing to the circularity of materials. *Procedia CIRP* 2020;90:67–72. <https://doi.org/10.1016/j.procir.2020.02.052>.
- [60] Vogtlander JG, Bijma A, Brezet HC. Communicating the eco-efficiency of products and services by means of the eco-costs/value model. *J Clean Prod* 2002;10:57–67. [https://doi.org/10.1016/S0959-6526\(01\)00013-0](https://doi.org/10.1016/S0959-6526(01)00013-0).
- [61] Vercauteren A, Spirinckx C, Geerken T. Life cycle assessment and eco-efficiency analysis of drinking cups used at public events. *Int J Life Cycle Assess* 2010;15:221–30. <https://doi.org/10.1007/s11367-009-0143-z>.
- [62] Pérez-Belis V, Bakker C, Juan P, Bovea MD. Environmental performance of alternative end-of-life scenarios for electrical and electronic equipment: a case study for vacuum cleaners. *J Clean Prod* 2017;159:158–70. <https://doi.org/10.1016/j.jclepro.2017.05.032>.
- [63] Severis RM, Simioni FJ, Moreira JMMAP, Alvarenga RAF. Sustainable consumption in mobility from a life cycle assessment perspective. *J Clean Prod* 2019;234:579–87. <https://doi.org/10.1016/j.jclepro.2019.06.203>.
- [64] Nurhadi H, Borén S, Ny H, Larsson T. Competitiveness and sustainability effects of cars and their business models in Swedish small town regions. *J Clean Prod* 2017;140:333–48. <https://doi.org/10.1016/j.jclepro.2016.04.045>.
- [65] Sigüenza CP, Cucurachi S, Tukker A. Circular business models of washing machines in The Netherlands: material and climate change implications toward 2050. *Sustain Prod Consum* 2021;26:1084–98. <https://doi.org/10.1016/j.spc.2021.01.011>.
- [66] De Almeida ST, Borsato M. Extending the RIPEX exergy-based method for selecting End of Life strategy. *Resour Conserv Recycl* 2020;152:104536. <https://doi.org/10.1016/j.resconrec.2019.104536>.
- [67] Foteinis S. How small daily choices play a huge role in climate change: the disposable paper cup environmental ban. *J Clean Prod* 2020;255:120294. <https://doi.org/10.1016/j.jclepro.2020.120294>.
- [68] Makov T, Meylan G, Powell JT, Shepon A. Better than bottled water?—energy and climate change impacts of on-the-go drinking water stations. *Resour Conserv Recycl* 2019;143:320–8. <https://doi.org/10.1016/j.resconrec.2016.11.010>.
- [69] Gallego-Schmid A, Jeswani HK, Mendoza JMF, Azapagic A. Life cycle environmental evaluation of kettles: recommendations for the development of eco-design regulations in the European Union. *Sci Total Environ* 2018;625:135–46. <https://doi.org/10.1016/j.scitotenv.2017.12.262>.
- [70] Changwichean K, Gheewala SH. Choice of materials for takeaway beverage cups towards a circular economy. *Sustain Prod Consum* 2020;22:34–44. <https://doi.org/10.1016/j.spc.2020.02.004>.
- [71] Kawaguchi T, Murata H, Fukushige S, Kobayashi H. Scenario analysis of car- and ride-sharing services based on life cycle simulation. *Procedia CIRP* 2019;80:328–33. <https://doi.org/10.1016/j.procir.2019.01.051>.
- [72] Spangher L, Gorman W, Bauer G, Xu Y, Atkinson C. Quantifying the impact of U.S. electric vehicle sales on light-duty vehicle fleet CO2 emissions using a novel agent-based simulation. *Transp Res D Trans Environ* 2019;72:358–77. <https://doi.org/10.1016/j.trd.2019.05.004>.
- [73] Nessi S, Rigamonti L, Grosso M. LCA of waste prevention activities: a case study for drinking water in Italy. *J Environ Manag* 2012;108:73–83. <https://doi.org/10.1016/j.jenvman.2012.04.025>.

- [74] Haapala KR, Brown KL, Sutherland JW. A life cycle environmental and economic comparison of clothes washing product-service systems. *Trans North Am Manuf Res Inst SME* 2008;36:333–40.
- [75] Kumazawa T, Kobayashi H. A simulation system to support the establishment of circulated business. *Adv Eng Inf* 2006;20:127–36. <https://doi.org/10.1016/j.aei.2005.11.004>.
- [76] Martin M, Lazarevic D, Gullström C. Assessing the environmental potential of collaborative consumption: Peer-to-peer product sharing in Hammarby Sjöstad, Sweden. *Sustainability* 2019;11:190. <https://doi.org/10.3390/su11010190>.
- [77] Amatuni L, Ottelin J, Steubing B, Mogollón JM. Does car sharing reduce greenhouse gas emissions? Assessing the modal shift and lifetime shift rebound effects from a life cycle perspective. *J Clean Prod* 2020;266:121869. <https://doi.org/10.1016/j.jclepro.2020.121869>.
- [78] Piontek FM, Amasawa E, Kimita K. Environmental implication of casual wear rental services: case of Japan and Germany. *Procedia CIRP* 2020;90:724–9. <https://doi.org/10.1016/j.procir.2020.01.076>.
- [79] Martin M, Heiska M, Björklund A. Environmental assessment of a product-service system for renting electric-powered tools. *J Clean Prod* 2021;281:125245. <https://doi.org/10.1016/j.jclepro.2020.125245>.
- [80] Allais R, Gobert J. Environmental assessment of PSS, feedback on 2 years of experimentation. *Matériaux Tech* 2018;105:504. <https://doi.org/10.1051/mattech/2018010>.
- [81] Weber CL, Koomey JG, Matthews HS. The energy and climate change implications of different music delivery methods. *J Ind Ecol* 2010;14:754–69. <https://doi.org/10.1111/j.1530-9290.2010.00269.x>.
- [82] Sivaraman D, Pacca S, Mueller K, Lin J. Comparative energy, environmental, and economic analysis of traditional and E-commerce DVD rental networks. *J Ind Ecol* 2007;11:77–91. <https://doi.org/10.1162/jiec.2007.1240>.
- [83] Fernando C, Soo VK, Doolan M. Life cycle assessment for servitization: a case study on current mobility services. *Procedia Manuf* 2020;43:72–9. <https://doi.org/10.1016/j.promfg.2020.02.112>.
- [84] Glew D, Stringer LC, Acquaye AA, McQueen-Mason S. How do end of life scenarios influence the environmental impact of product supply chains? comparing biomaterial and petrochemical products. *J Clean Prod* 2012;29–30: 122–31. <https://doi.org/10.1016/j.jclepro.2012.02.005>.
- [85] Huijbregts MAJ, Steinmann ZJN, Elshout PMF, Stam G, Verones F, Vieira M, et al. ReCiPe2016: a harmonised life cycle impact assessment method at midpoint and endpoint level. *Int J Life Cycle Assess* 2017;22:138–47. <https://doi.org/10.1007/s11367-016-1246-y>.
- [86] Goedkoop M, Spriensma R. The eco-indicator 99: a damage oriented method for life cycle impact assessment. Methodology report. PRÉ Consultants 2001.
- [87] Hirschler R, Böni HW. Combining environmental and economic factors to evaluate the reuse of electrical and electronic equipment – a Swiss case study. *Resour Conserv Recycl* 2021;166:105307. <https://doi.org/10.1016/j.resconrec.2020.105307>.
- [88] Moreau H, de Meux L de J, Zeller V, D'Ans P, Ruwet C, Achten WMJ. Dockless e-scooter: a green solution for mobility? Comparative case study between dockless e-scooters, displaced transport, and personal e-scooters. *Sustainability* 2020;12: 1803. <https://doi.org/10.3390/su12051803>.
- [89] Simon B, Amor MB, Földényi R. Life cycle impact assessment of beverage packaging systems: focus on the collection of post-consumer bottles. *J Clean Prod* 2016;112:238–48. <https://doi.org/10.1016/j.jclepro.2015.06.008>.
- [90] Potting J, van der Harst E. Facility arrangements and the environmental performance of disposable and reusable cups. *Int J Life Cycle Assess* 2015;20: 1143–54. <https://doi.org/10.1007/s11367-015-0914-7>.
- [91] Ding N, Pan J, Zhang Z, Yang J. Life cycle assessment of car sharing models and the effect on GWP of urban transportation: a case study of Beijing. *Sci Total Environ* 2019;688:1137–44. <https://doi.org/10.1016/j.scitotenv.2019.06.111>.
- [92] Boldoczki S, Thorenz A, Tuma A. The environmental impacts of preparation for reuse: a case study of WEEE reuse in Germany. *J Clean Prod* 2020;252:119736. <https://doi.org/10.1016/j.jclepro.2019.119736>.
- [93] Sun S, Ertz M. Environmental impact of mutualized mobility: evidence from a life cycle perspective. *Sci Total Environ* 2021;772:145014. <https://doi.org/10.1016/j.scitotenv.2021.145014>.
- [94] Fernando C, Soo VK, Compston P, Kim HC, de Kleiner R, Weigl D, et al. Life cycle environmental assessment of a transition to mobility servitization. *Procedia CIRP* 2020;90:238–43. <https://doi.org/10.1016/j.procir.2020.01.098>.
- [95] Amasawa E, Shibata T, Sugiyama H, Hirao M. Environmental potential of reusing, renting, and sharing consumer products: systematic analysis approach. *J Clean Prod* 2020;242:118487. <https://doi.org/10.1016/j.jclepro.2019.118487>.
- [96] Bonilla-Alicia RJ, Watson BC, Shen Z, Tamayo L, Telenko C. Life cycle assessment to quantify the impact of technology improvements in bike-sharing systems. *J Ind Ecol* 2020;24:138–48. <https://doi.org/10.1111/jiec.12860>.
- [97] Severengiz S, Finke S, Schelte N, Forrister H. Assessing the environmental impact of novel mobility services using shared electric scooters as an example. *Procedia Manuf* 2020;43:80–7. <https://doi.org/10.1016/j.promfg.2020.02.114>.
- [98] Mont O. Reducing life-cycle environmental impacts through systems of joint use. *Greener Manag Int* 2004;45:63–77.
- [99] Zamani B, Sandin G, Peters GM. Life cycle assessment of clothing libraries: can collaborative consumption reduce the environmental impact of fast fashion? *J Clean Prod* 2017;162:1368–75. <https://doi.org/10.1016/j.jclepro.2017.06.128>.
- [100] Hoffmann BS, de Simone Morais J, Teodoro PF. Life cycle assessment of innovative circular business models for modern cloth diapers. *J Clean Prod* 2020; 249:119364. <https://doi.org/10.1016/j.jclepro.2019.119364>.
- [101] Williams ED, Sasaki Y. Energy analysis of end-of-life options for personal computers: resell, upgrade, recycle. *Proc IEEE Int Symp Electron Environ* 2003: 187–92. <https://doi.org/10.1109/isee.2003.1208072>.
- [102] Krystofik M, Babbitt CW, Gaustad G. When consumer behavior dictates life cycle performance beyond the use phase: case study of inkjet cartridge end-of-life management. *Int J Life Cycle Assess* 2014;19:1129–45. <https://doi.org/10.1007/s11367-014-0713-6>.
- [103] Intlekofer K, Bras B, Ferguson M. Energy implications of product leasing. *Environ Sci Technol* 2010;44:4409–15. <https://doi.org/10.1021/es9036836>.
- [104] Intlekofer K, Bras B, Ferguson M. Reducing product energy consumption with leasing. *ASME Des Eng Tech Conf* 2009;275–89. <https://doi.org/10.1115/DETC2009-87328>.
- [105] Amaya J, Lelah A, Zwolinski P. Design for intensified use in product-service systems using life-cycle analysis. *J Eng Des* 2014;25:280–302. <https://doi.org/10.1080/09544828.2014.974523>.
- [106] Severengiz S, Schelte N, Bracke S. Analysis of the environmental impact of e-scooter sharing services considering product reliability characteristics and durability. *Procedia CIRP* 2020;96:181–8. <https://doi.org/10.1016/j.procir.2021.01.072>.
- [107] Severengiz S, Finke S, Schelte N, Wendt N. Life cycle assessment on the mobility service E-scooter sharing. *IEEE Eur Technol Eng Manag Summit*; 2020. 10.1109/E-TEMS46250.2020.9111817.
- [108] Ahamed A, Vallam P, Iyer NS, Veksha A, Bobacka J, Lisak G. Life cycle assessment of plastic grocery bags and their alternatives in cities with confined waste management structure: a Singapore case study. *J Clean Prod* 2021;278:123956. <https://doi.org/10.1016/j.jclepro.2020.123956>.
- [109] Cordella M, Alfieri F, Sanfelix J. Reducing the carbon footprint of ICT products through material efficiency strategies: a life cycle analysis of smartphones. *J Ind Ecol* 2021;25:448–64. <https://doi.org/10.1111/jiec.13119>.
- [110] Baxter J. Systematic environmental assessment of end-of-life pathways for domestic refrigerators. *J Clean Prod* 2019;208:612–20. <https://doi.org/10.1016/j.jclepro.2018.10.173>.
- [111] Woods L, Bakshi BR. Reusable vs. disposable cups revisited: guidance in life cycle comparisons addressing scenario, model, and parameter uncertainties for the US consumer. *Int J Life Cycle Assess* 2014;19:931–40. <https://doi.org/10.1007/s11367-013-0697-7>.
- [112] Richter JL, Tähkämo L, Dalhammar C. Trade-offs with longer lifetimes? The case of LED lamps considering product development and energy contexts. *J Clean Prod* 2019;226:195–209. <https://doi.org/10.1016/j.jclepro.2019.03.331>.
- [113] PricewaterhouseCoopers. The sharing economy: consumer intelligence series. 2015.
- [114] Van Ostaeyen J, Van Horenbeek A, Pintelon L, Duflou JR. A refined typology of product-service systems based on functional hierarchy modeling. *J Clean Prod* 2013;51:261–76. <https://doi.org/10.1016/j.jclepro.2013.01.036>.
- [115] Weidema B, Wenzel H, Petersen C, Hansen K. The product, functional unit and reference flows in LCA. Danish Ministry of the Environment. *Environmental News* 2004;70.
- [116] Moraga G, Huysveld S, Mathieux F, Blengini GA, Alaerts L, Van Acker K, et al. Circular economy indicators: what do they measure? *Resour Conserv Recycl* 2019;146:452–61. <https://doi.org/10.1016/j.resconrec.2019.03.045>.
- [117] Ellen MacArthur Foundation, Granta Design. Circularity Indicators: an approach to measuring circularity. Methodology 2015.
- [118] Linder M, Sarasini S, van Loon P. A metric for quantifying product-level circularity. *J Ind Ecol* 2017;21:545–58. <https://doi.org/10.1111/jiec.12552>.
- [119] Pauliuk S. Critical appraisal of the circular economy standard BS 8001:2017 and a dashboard of quantitative system indicators for its implementation in organizations. *Resour Conserv Recycl* 2018;129:81–92. <https://doi.org/10.1016/j.resconrec.2017.10.019>.
- [120] Cambridge Econometrics, Trinomics, ICF. Impacts of circular economy policies on the labour market. Final report and annexes. European Union; 2018.
- [121] Groen EA, Bokkers EAM, Heijungs R, de Boer IJM. Methods for global sensitivity analysis in life cycle assessment. *Int J Life Cycle Assess* 2017;22:1125–37. <https://doi.org/10.1007/s11367-016-1217-3>.
- [122] Wei W, Larrey-Lassalle P, Faure T, Dumoulin N, Roux P, Mathias JD. How to conduct a proper sensitivity analysis in life cycle assessment: taking into account correlations within LCI data and interactions within the LCA calculation model. *Environ Sci Technol* 2015;49:377–85. <https://doi.org/10.1021/es502128k>.
- [123] Södersten CJ, Wood R, Hertwich EG. Environmental impacts of capital formation. *J Ind Ecol* 2018;22:55–67. <https://doi.org/10.1111/jiec.12532>.
- [124] Södersten CJ, Wood R, Wiedmann T. The capital load of global material footprints. *Resour Conserv Recycl* 2020;158:104811. <https://doi.org/10.1016/j.resconrec.2020.104811>.
- [125] Vivanco DF. The role of services and capital in footprint modelling. *Int J Life Cycle Assess* 2020;25:280–93. <https://doi.org/10.1007/s11367-019-01687-7>.
- [126] Koide R, Lettenmeier M, Kojima S, Toivio V, Amellina A, Akenji L. Carbon footprints and consumer lifestyles: an analysis of lifestyle factors and gap analysis by consumer segment in Japan. *Sustainability* 2019;11:5983. <https://doi.org/10.3390/su11215983>.
- [127] Koide R, Kojima S, Nansai K, Lettenmeier M, Asakawa K, Liu C, et al. Exploring carbon footprint reduction pathways through urban lifestyle changes: a practical approach applied to Japanese cities. *Environ Res Lett* 2021;16:084001. <https://doi.org/10.1088/1748-9326/ac0e64>.
- [128] Koide R, Lettenmeier M, Akenji L, Toivio V, Amellina A, Khodke A, et al. Lifestyle carbon footprints and changes in lifestyles to limit global warming to 1.5 °C, and ways forward for related research. *Sustain Sci* 2021;16:2087–99. <https://doi.org/10.1007/s11625-021-01018-6>.